

FOREIGN EXCHANGE FORECASTING
USING LSTM MACHINE MODEL

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FOREIGN EXCHANGE FORECASTING USING LSTM MACHINE MODEL

CHENG YUN

A project report submitted in partial fulfilment of the
requirements for the award of the degree of
Master in Data Science

Faculty of Computing
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JANUARY 2026

DECLARATION

I declare that this project report entitled “*Foreign Exchange Forecasting Using LSTM Machine Model*” is the result of my own research except as cited in the references. The project report has not been accepted for any degree and is not concurrently submitted in candidature of any other degree.

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ACKNOWLEDGEMENT

While writing this thesis,I read tons of relative literature.I am genuinely grateful for those authors who provide so many delicate research papers and books.I am also thankful to my honorable supervisor,Dr. ROLIANA BINTI IBRAHIM .She offers me some valuable and significant suggestions for boosting the quality of my thesis.Without her continued support and interest, this thesis may not be presented here.

I am also thankful to my parents and sister.During the term of my thesis ,they really gave me huge support to complete my thesis.

ABSTRACT

Curently,the market of foreign exchange rates is biggest market.Everyday,there is almost 7 trillion dollars trading in forex market.Massive market attracts mulitiple skatesholder such as bankers,central bank,individual investors.However,this market is quite sensitive,and any crisis can affact it significantly.For example,because of Ukraine-Russia conflicts,Russia official currency: Rouble met a damtical decreasing against USD.Futhermore,because of repeatly tarriff war between China and US,it results the unstable fluctuation between CNY and USD.According to the 2022 financial reports of Google and Apple, they lost approximately \$1.511 billion USD due to dramatic exchange rate movements. Nowadays,trariff wars already spead its magic in forex market.It is a crucial goal to predict the real-time and short-term futurn value of exchange rate accurately.For the extreme requirment of accuracy prediction of forex,machine model field is also developping day by day. There are some important macroeconomic indictors should be considered,including interest,CPI and long-term government debt rate.Those indicators can comprehensively displays the picture of country's basic economy.In macroeconomic thoery,those three indicators most influence the exchange rate . More and more new and advanced architecture come out ,just for predict the value of forex accurately like RNN,GRU,ARIMA,Transformer and so on. Currently,in those models,LSTM performs loudly.LSTM can capture long-term Dependencies,operate time-series forecasting well,handle non-linear dynamics,and even with uncleaned data,it still performs well.Therefore,in this study,it major develops a machine learning model to forcase the future value of exchange rate based on LSTM architecture.It also retrieved historical data from interest ,CPI and long-term government debt rate. It can seen that machine learning models play a more and more important role in forex considering rencent literatures.

ABSTRAK

Pada masa ini, pasaran kadar pertukaran asing merupakan pasaran terbesar di dunia. Setiap hari, hampir 7 trilion dolar Amerika didagangkan dalam pasaran forex. Pasaran yang besar ini menarik pelbagai pemain seperti bank, bank pusat, dan pelabur individu. Walau bagaimanapun, pasaran ini agak sensitif, dan sebarang krisis boleh menjejaskannya dengan ketara. Sebagai contoh, akibat konflik Ukraine-Russia, mata wang rasmi Rusia: Ruble mengalami penurunan mendadak berbanding USD. Tambahan pula, disebabkan perang tarif berulang antara China dan AS, ia mengakibatkan turun naik yang tidak stabil antara CNY dan USD. Menurut laporan kewangan 2022 Google dan Apple, mereka kehilangan kira-kira \$1.511 bilion USD akibat pergerakan kadar pertukaran yang mendadak. Kini, perang tarif telah menular ke pasaran forex. Adalah matlamat penting untuk meramal nilai masa nyata dan jangka pendek kadar pertukaran dengan tepat. Bagi keperluan ramalan forex yang sangat tepat, bidang model mesin juga berkembang dari hari ke hari. Terdapat beberapa penunjuk makroekonomi penting yang perlu dipertimbangkan, termasuk kadar faedah, CPI dan kadar hutang kerajaan jangka panjang. Penunjuk-penunjuk ini dapat memaparkan gambaran ekonomi asas sesebuah negara secara menyeluruh. Dalam teori makroekonomi, ketiga-tiga penunjuk ini paling mempengaruhi kadar pertukaran. Semakin banyak seni bina baru dan maju muncul, hanya untuk meramal nilai forex dengan tepat seperti RNN, GRU, ARIMA, Transformer dan sebagainya. Pada masa ini, dalam model-model tersebut, LSTM menunjukkan prestasi yang memberangsangkan. LSTM dapat menangkap kebergantungan jangka panjang, mengendalikan ramalan siri masa dengan baik, mengurus dinamik tidak linear, dan walaupun dengan data yang tidak bersih, ia masih berprestasi baik. Oleh itu, dalam kajian ini, ia terutama membangunkan model pembelajaran mesin untuk meramal nilai masa depan kadar pertukaran berdasarkan seni bina LSTM. Ia juga mengambil data sejarah daripada kadar faedah, CPI dan kadar hutang kerajaan jangka panjang. Dapat dilihat bahawa model pembelajaran mesin memainkan peranan yang semakin penting dalam forex berdasarkan literatur terkini.

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LIST OF ABBREVIATIONS

LSTM	-	Long-term short memory
CPI	-	Consumer Price Index
US	-	the United States
USD	-	United States Dollars
CNY	-	Chinese Renmingbi
FOREX	-	Foreign Exchange Rate
RNN	-	Recurrent Neural Network
RNA	-	Single-stranded Nucleic Acid
RMSE	-	Root Mean Squared Error
MAE	-	Mean Absolute Error
MAPE	-	Mean Absolute Percentage Error
US	-	The United States
VECM	-	Vector Error Correction Model Method
ADF	-	Augmented Dickey-Fuller

CHAPTER 1

INTRODUCTION

1.1 Overview

Through the whole financial market across the world, it is the fact that Foreign exchange market occupies quite an enormous part of the financial market pie. Many stakeholders such as central bankers, transnational enterprises and even individual investors, try best to predict the trending of FX in the future.

However, due to the dynamic, complex variables, sensitivity of the FX market, it is a big challenge to make a successful and valued decision for investing. For extracting extra profit through FX ocean, developing advanced machine algorithms could be a significant break-through to forecast the trending of the foreign exchange market.

Recently, Long Short Term Memory (LSTM) has been used as a smart tool to address the problems of the stakeholders' suspense. LSTM can offer a powerful method to analyse the time-series problem while processing the historical data. Thus, LSTM already is applied into different disciplines for outputting a successful decision-making.

In this project, it demonstrates a new perspective to tackle the FX predicting concerns with the help of the LSTM. From data collection, data preprocessing, model training, model tuning to model prediction, all steps are reflected in this project to convey a new side for stakeholders.

1.2 Problem Background

Every day, the FX market involves around 9.6 trillion dollars around the world, according to the report of BIS press (BIS, 2025). Because of the huge capitals, every fluctuation of the FX will involve thousands of dollars loss or creation. There are some cases to explain how FX influences decision-making among the government, the central bank and enterprises.

During 2020 to 2023, Sri Lanka met a huge national debt crisis, which destroyed the national economy and led to extreme unrest across the country. Due to unpaying large national debt, which was valued by foreign currencies, the rupee, the official currency of Sri Lanka, lost its value rapidly in short time. At the same time, because the rupee collapsed quickly, it contributes to the rise of debt value by foreign currencies defined further. As the consequence, it leads to a vicious circle on the country side. Till now, the economy of Sri Lanka is trying to recover from this terrible crisis. According to the World Bank dataset, now the GDP per capita of Sri Lanka was 4.46 thousand USD in 2019, compared to 2025, this one reaches 4.19 thousand USD. (World Bank, 2025) It suggests the cruel consequences of that Sri Lanka manages FX badly.

From 2018 to 2025, Turkey's lira has been losing real value because TCMB (the central bank of the Republic of Turkey) lost the control of CPI. At the first year of 2018, while Turkey met high inflation, TCMB should improve the interest to reduce the increasing of CPI. However, TCMB adopted a low interest rate policy, which is against the basic economic theory, as the consequence of the loss of Lira rapidly. It also reflects the trending of Lira against USD, which changes from 1:4 (USD/Lira) in 2018 to 1:42.5 (USD/Lira) in 2025. Furthermore, TCMB has encountered substantial loss around years in the Foreign exchange market.

Reviewing the 2024 annual report of Coca Cola, it illustrated that the average of all operating currencies weakened 5% compared to 2023 against USD. In the lists of operating currencies, British pound strengthened 3% against USD, while CNY weakened 3% against USD compared to 2023. Without adopting FX strategy, it would affect the net revenue of the company. However, Coca Cola applies a set of hedging

activities to mitigate the effect of the fluctuation of foreign currencies. According to the report, it decreased 11% of the total impact of the foreign exchange rate fluctuation on net revenue.

1.3 Problem Statement

Since the FX is the biggest financial market, multiple machine learning models are applied in this big cake area. Although there are set of machine learning models which research various sentiments that may impact on FX market, in this study, it majorly talks about the macroeconomic indicators influencing the FX market of CNY and USD.

LSTM can process the historical data of exchange rate between CNY and USD perfectly and efficiently to predict the future trending of CNY and USD. Based on current research, it can be told that LSTM performs accurately and stably on processing time-series data because of the specialities such as selectively remember and forget.

However, few studies discuss the influence of integrating CPI, interest and long term government debt. Thus, this project is dedicated to training a model to address the accuracy of predicting FX based on the algorithm framework of LSTM.

1.4 Research Question

- (a) How to retrieve financial data from multiple platforms for integrating a valued dataset for training and test phrases?
- (b) How to improve the accuracy of predicting the exchange rate between CNY and USD based on LSTM ?
- (c) How to transform raw results to tangible conclusions by using Matplotlib?

1.5 Research Aims

The aim of this project is to retrieve data from different platforms, and process the data for training, testing and tuning under the LSTM algorithm, for outputting a well-performing model to forecast the future value of the exchange rate between CNY and USD.

1.6 Research Objectives

The research objectives show below items:

- (a) To retrieve the raw data from central bank websites and a financial website(Alpha Advantage), and process these data for modelling .
- (b) To train, test and tune model for improving performance and decreasing errors, meantime output a high-accurate forecasting value of the exchange rate between CNY and USD.
- (c) To develop a friendly-using and effective dashboard summarising the total raw result from the model for offering a tangible and valued insight for stakeholders.

1.7 Project Scope

Due to the essential factors of the machine learning model, this project collects the data, interest and 10-Y bonds from 2014 to 2024, which is from the central bank databases and market platforms. The fundamental model focuses on predicting real-time and next-day price trending, rather than the long-term forecasting.

As for machine learning model implementations, it only focuses on the LSTM expiration. This project does not adopt hybrid or multiple models to reinforce the

ability of the forecasting model framework. It also excludes multiple macroeconomic sentiment considerations.

In this project, in order to evaluate and develop the machine learning model for predicting the price of the forex market, it is limited to analyse multiple currency pairs. However, it only targets currency pairs of USD/CNY.

In terms of academics, this project can provide a new perspective to develop and promote the world of the forex forecasting model. It also can be sure to offer a practical for future research involving forex predicting models.

1.8 Project Significance

This project is expected to be dedicated to providing a new angle for solving the problem which can not accurately foresee the price of forex in a short time, and develop a new method which involves mastering a machine learning model for forecasting the short-term price of currency pairs in the forex market for prediction.

For investors, real-time and next-day price of forex can significantly influence the final decision if increasing the investment for gaining extra interest or purchasing a hedging investment portfolio for the risk of forex loss. On the other hand, as for central banks, it is a crucial mission to maintain the stability of currency effectively. Real-time and next-day prediction can offer these facilities a relative reference when they are making policies for combating violently fluctuations of currency in market, and ensure oscillation mildly.

In terms of academics, this project can provide a new perspective to develop and promote the world of the forex forecasting model. It also can be sure to offer a practical for future research involving forex predicting models.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

The FOREX financial market can be confirmed as the most variable and biggest market around the world. Although there are many methods and models to analyse and forecast the FOREX market. Because of the unpredictability and sensitivity of its characteristics for macroeconomic factors, it is significantly challenging to apply accurate forecasting of the future values. However, with its intensive time-series relevance, recently, there are developing loads of advanced techniques of ML (Machine learning), which erect more steadfast and precise predictive models such as LSTM (Long Short-Term Memory), to address the accuracy issue of the FOREX predicting. Through the overall literature, it demonstrates that identifying the key features about the FOREX such as CPI (consumer price indices), interest rate and 10-year bond rate. To visualize the trending of currency pairs, it is necessary to adopt a suitable tool to draw it.

2.2 Forex Significance

As the biggest and most valued market in the world, the forex market always plays a crucial role in the financial account of the company, national foreign exchange reserves of the country and even the assets of individuals. It also pushes these stakeholders to mitigate the risk of forex and make a positive reaction.

According to the report of THE EDG Malaysia (2024, THE EDG Malaysia), it said a company called Kek Seng (M) Bhd which fell into red due to the forex loss. From

the financial statement of Kek Seng (M) Bhd, it revealed a net loss of RM36.66 million in the third quarter ended Sept 30, 2024 because it was dragged down by unrealised forex losses amounting to a massive RM81.84 million. However, the 2024 annual report of Apple (2024, Annual Report.com) displayed that the Apple applied a value-at-risk (“VAR”) model to assess the potential impact of fluctuations in exchange rates. Based on the results of the model, the Company estimates, with 95% confidence, a maximum one-day loss in fair value of \$538 million till September 28, 2024. With the early warning of the model, the Apple adapt a series of measurements to hedge the risk of forex such as using derivative instruments, offset exposures, or adjust local currency pricing of its products and services.

Reuters reported that The National Bank of Poland (NBP) claimed the loss of exchange rate differences on the 2023 result amounted to 31.0 billion zlotys facing the COVID-19 pandemic. (2024, Reuters) It massively affected the amount of foreign currency reserves negatively and decrease the stability of the zlotys in forex market. In the contrast, Argentinian central bank announced that it will extend their long-standing \$18 billion bilateral currency swap for another year with Chinese central bank on April, 2024. (2024, AP) It can give a breath room for the current depleting foreign currency reserves rather than using dollar to trade. Swapping deals can ensure the constancy of the trade between Argentina and China facing the risk of forex against USD.

2.3 Data Parameter

Macroeconomic indicators play a crucial role in foreign exchange rate movement, such as CPI, interest and 10-Year bond. There are some cases to interpret the relationship among these factors. For an instance, Kim and his mates (Kim, et al, 2022) fathered the inadequacy of traditional econometric models, such as the uncovered interest parity (UIP) and purchasing power parity (PPP), which exceed the random basic models. These indicators can be confirmed as the key dataset for machine learning processing to forecast the future foreign exchange rate.

2.3.1 Interest

Interest is the price paid for borrowing money. It is usually described as the percentage rate around a certain short-term period. Although there are two different interest rates: normal interest and the real interest rate. Dr. Mankiw defined that normal interest is the bank paid, and the real interest is the increase in your purchasing power. (2010, N. Gregory Mankiw). In this project, it is talking about the factor of influencing the forex under the normal interest which is from the central bank.

Interest can be observed as a significant reference point which affects the fluctuation of the forex. Some studies have interpreted how interest rate movements impact on exchange rate over the time. For an instance, as the display of Yuenan (2023)'s study, while a central bank of the country increases their interest such as the Federal Reserve in the United States, it will lead to the result that rising their currency, because international investments will benefit from high interest level, and then in the financial market, the demand of purchasing USD regularly rise. This study also declaim that because of interest highly boosting, the Chinese Yuan has devalued significantly. Through his study which discussed and implemented the impact of Federal Reserve System raising interest rate on the exchange rate of the US Dollar against the Chinese Yuan, with the analysis of ARIMA, it illustrated that hiking the interest rate of the Federal Reserve led to the exchange rate of pairs USD/CNY steadfast increased over the time.

In addition, according to Marcin Kolasa's study (Marcin Kolasa, et, 2022), they added behavioral agents to stretch the standard open economy New Keynesian framework. It conveyed a theory that the exposing to interest rate parity condition in a way and consisting with recent testing-evidence reconsidered that conducting positive monetary policy such as boosting the interest by central bank can significantly influence international output comovement positively. Moreover, they developed the New Keynesian model with appraising an extension to successfully mitigated many problems which demonstrated the relationship between exchange rates and interest rates. Through their improvement of the model, they observed that the trending of the exchange rate under two different monetary policy statuses: Conventional Monetary

Shocks and Low for Longer Policies. In the final exhibition, the result showed that during these statuses, the trending of exchange rate was intensively affected by the interest movement.

2.3.2 Consumer Price Index

Consumer price index is a measure of the level of the price, consisting of the prices of thousands of goods and services. Also it can clearly reveal the pricing level of a nation and be considered as a barometer of the country's economy and interest of the central bank.

The CPI provides effective insight into how inflationary pressures may impact market patterns, company financial tables, and stakeholders. Joseph and his mate (Joseph, et 2023) researched the relationship between exchange rate and consumer price index with data context from South African. They adopt OLS (ordinary least squares) regression with the aid of Gretl econometric and statistics software to analyse the South Africa 2022 Consumer Price Index (CPI) YoY. After modeling, they found that two-month later, there are a possible and positive movement in exchange rate while consumer price index get alteration. Furthermore, offering policies of economy and finance that may cushion the exchange rate volatility could catalyse a favourable effect on domestic prices, which is direly needed for the economic good of the citizens. They summarized their project and concluded that the oscillation of South African currency may positively and significantly impact the consumer price index, which means that there is a powerful relevance between CPI and FOREX.

There is another study to demonstrate the powerful relationship between CPI and foreign exchange rate. Through the study of Didarul and his mates (Didarul, et 2024), it is not herculean to recognise the dynamics of Uruguay's real effective exchange rate with the aid of the MundellFleming model. They analyzed crucial economic indicators and employed econometric techniques, and therefore revealed the significant insights of the determinants of Uruguay's currency. With the analysis of the short-term real effective exchange rate in Uruguay currency, they deployed the

Ordinary Least Squares(OLS) regression model.After processing,they concluded that with maintained appropriate CPI,it will promote Uruguay currency healthily.

2.3.3 Long-term Government Rate

10 years national debt is the interest rate the government pays to borrow money for a decade.It can be recognised by a benchmark which refers to the other interest rates.The investors often concerns it as a key indicator of the economic condition.

As the study of Rosnawintang and his mates(2019,Rosnawintang,et),they analyzed the relationship between Indonesian currency and bond yield.According to their study ,it is a crucial theory that domestic interest rate is the sum of international interest rate and expectation of depreciation (or appreciation) of domestic currency exchange rate against foreign currency .Through their VAR(Vector autoregression) model,they collected the data of IDR/USD exchange rate and bond yield from January 2006 to December 2018 with monthly time series.After the feature engineering,they concluded that it is effective that the causal relationship between the government bond yield and paired IDR and USD exchange rate during the short-term relationship.They also declaimed that in the first three months, the response of the government bond yield against the exchange rate between IDR and USD was very strong (significant 1%).

Moreover, Litsios, et al,(2019) researched what role national debt played in exchange rate between US dollar and Yen.They implemented an alternative approach to decide national debt plays a clear role in long-term exchange rate with the basis of micro-foundations .They adopted their theoretical model, Vector Error Correction Model (VECM), to test data from USA(treated as the foreign economy) and Japan(treated as the domestic economy),including nominal exchange rates between USD and Yen from 1983 to 2015,nominal money supply(M3) of USA and Japan and three month rate on Treasury of Japan and USA.They found that comparing current literatures which ignores government debt as a key determinant of long-term exchange rate , on the contrary,they emphasize the importance of government debt for long-term

exchange rate heavily. Furthermore, they stated that an increasing government debt led to a depreciation of the yen against the dollar in the long-term while an increasing of the relative foreign (USA) variables led to a nominal yen appreciation in the long period.

2.4 Current Applied Approaches to Forex Prediction

Nowaday, there are many extraordinary approaches to predict the forex and solve the accuracy of forex prediction. However, there are plenty of models that only concern the exchange rate itself or just part of three microeconomic sentiments (CPI, interest and 10 years government debt). Because of limitation of factors input, the accuracy of forecasting is relatively low. Although there is exiting some significant shortages, models applied and data processing still can contribute to a new perspective. In this sub-section, there are seven cases to discuss.

Jamali, et al, (2023) developed a hybrid model to reinforce the availability of predicting forex. During the research, they adopted multiple regression and simulated annealing to process the data. Their dataset only included the exchange rate of EUR/USD pairs from 25, October 2016 to 27 September 20, 2020. Due to the ability of detecting relative influence of one or more predictor variables and simplifying the programme of calculation and interpretation, they decided to apply multiple regression for analyzing the exchange rate data. They also noticed that using simulated annealing model can optimize the currency prediction results which were generated by multiple regression. They stated that two techniques combination models can improve the accuracy of the result of prediction.

Moustafa, et al (2023) constructed two different framework models to process data for prediction. At same time, they compared the result of two different models generated for checking which one can perform better. Considering the dataset they were collecting, the limitation of data diversity was observed. In their dataset, they only collected the historical data of exchange rate. In other hand, comparing two different architecture of models can significantly devoted to a new view with concerning model

usage during the prediction. The concluded that FNF-CNN combination performed much better than baseline combination(FNF-SVR), and FNF can generate variety of features for decreasing the error meantime CNN can offer efficient feature learning.

Kamalov, et al(2022) implemented a new method ,outlier detection method(PKDE) in the forex field for demonstrating it overperformed than others relative models. They compared with five different models while they set the proposed model, PKDE. They only focus on historical exchange rate data from 1 January ,1999 to 1 September,2019, although they collected multiple currency pairs to ensure stability of objective. Because of the limitation of factors, it may lead to low accuracy result. After processing same data in different models, they concluded that PKDE obviously outperformed other machine learning approaches in forecasting of forex. They recognised that it is a shortage to predict the continue target.

Meng, et al(2024) explored multiple advanced deep learning models for leveraging the accuracy of exchange rate ,specially about CNY/USD pairs. In this research, they collected the variety data such as macroeconomic statistics like trade value (import and export) , short-term capital flows and so on, and financial market trading Data, including stock indices and exchange rates. Comparing with early literature, they substantially promote the variety of data . However, the time series of these data only include from May,2019 to May,2020. Because of limitation of time, it may decrease the accuracy and reliability of models . After analysing data under the six different model, which including LSTM, TCN, TSMixer, TimesNet, MLPM, and FED, they stated that advanced architectures of TSMixer, FED, and iTransformer performed well than old ones. Also they also claimed that transformer-based models contribute to a significant improvement in financial forecasting area for gaining leverage of accuracy and reliability. They also admitted that limitation of currency choice ,short-term time collection and the challenge of GradCAM decision-making can partly affect the accuracy and reliability.

Alaminos,et al (2023) developed a new simulation for forex prediction by using Auxiliary-Field Quantum Monte Carlo (AFQMC) technique.They collected historical data of exchange rate of USD/EUR and USD/JAP from 2015 to 2021 to test the model ,and other crucial factors of macroeconomic indicators such as interest ,long term national debt and so on .They tested some major models with the help of AFQMC to facilitate the accuracy of these models.Comparing with existing literature and methods for forex forecasting,they concluded that AFQMC significantly leverage the accuracy and reliability of forex forecasting ,and displayed a low level of error and illustrated stability in appraising FOREX market models.They also considered highly accuracy of models can significantly positively influence the decision-making for stakeholders in forex prediction.

Luo,et al (2023) researched the evidence of how COVID-19 pandemic impacted the forex market.It offered a valued view to check breakdown issues or crises if it substantially influences the forex market.They collected the benchmark interest rate,investor sentiment and panic index for testing the models.Processing the data under different models,they found that RNN is the model overperformed other models in currency pairs USD/JPY and USD/CNY of forecasting , while LSTM is the best one in currency pairs GBP/USD of predicting in the COVID-19 pandemic period.In other hands,estimating the models performance,they made a conclusion that CNN and LSTM performs well than other traditional model in exchange rate forecasting ,specially concerning the sentiment factor ,COVID19 pandemic.

Overall,they contributed new perspectives to current researchers in forecasting the forex market field.However, there are still some sentiment variables they did not concern.Some of them only focus on exchange rate historic data.With relatively limitation,it may make a low accurate prediction result.Thus,in this project,it is necessary to consider crucial macroeconomic indicators and time choice including economic breakdown term.For more clearly comparison of different cases,checking them on table is a good way to observe.Below Table 2.1 summarise these six studies.

Table 2.1 Summary of Six Cases in Forecasting the Forex Market

Papers	Topic	Dataset	Models	Limitation
Jamali,et al,(2023)	Using multiple regression, simulated annealing to predict the right action to make a successful decision	EUR/USD pair from 25 October,2016 to 27 September 2020	multiple regression and simulated annealing	The variety of dataset
Moustafa,etal(2023)	Predict prices in advance by using different models	Daily prices of currency pairs GBP/USD, EUR/USD, USD/CHF, USD/JPY, AUD/USD, and USD/CAD from 2000 to 2019.	FNF combine with other machine learning models such as FNF-SVR-RBF, FNF-CNN and other five combination.	The variety of dataset
Kamalov,etal(2022)	Forecasting instances of significant fluctuations in currency exchange rates	Currency pairs USD/EUR,USD/GBP,USD/YEN, and USD/AUD from 1 January,1999 to 1 September ,2019	PKDE, OLS, NNR and other 7 models	The variety of dataset

Paper	Topic	Dataset	Models	Limitation
Meng,et al(2024)	enhancing and exploring the predictive accuracy of the RMB/USD exchange rate by using advanced deep learning models	Multiple indicators of Fundamental Data, Stock Index Data, Current Account, Capital Account, Currency Market, and Exchange Rates such as CPI, interest and so on	LSTM, TCN, TSMixer, TimesNet, PatchTST, iTransformer, Fedformer, Transformer and MLP	Only one currency pair(RMB/USD),short-term of historical data collection.
Alaminos,et al (2023)	Implementing Auxiliary-Field Quantum Monte Carlo to increase the accuracy of the FOREX markets models	Exchange rate of currency pairs EUR/USD,YEN/USD from 2015 to 2021	Auxiliary-Field Quantum Monte Carlo combine with other models	The variety of dataset
Luo,et al (2023)	Investigating the influence of the COVID-19 pandemic in forex market	Exchange rate of currency pairs USD/JPY, GBP/USD, and USD/CNY from 23 January, 2020 to 14 September, 2021, benchmark interest rate ,and the panic index.	LightGBM, RNN,and other four models	

2.5 LSTM Networks

Recurrent Neural Network (RNN) is a quite famous architecture, and is adopted and applied in diversity scenes for tackling harsh problems. With the limitation of RNN, which displays shortcomings of vanishing Gradient Problem, short-term memory, and slow training, Dr. Sepp Hochreiter and Dr. Jürgen Schmidhuber developed the Long Short-Term Memory to facilitate these shortage of RNN in 1997. Typically, LSTM is a type of RNN, which is designed for remembering long-term dependencies and forgetting irrelevant information.

With the fabulous ability of LSTM, it can help financial players to make a successful decision. Mroua, et al (2023) applied LSTM to predict several financial indicators under COVID-19 pandemic crisis term. Firstly, they collected multiple historical financial time-series indicators from 2 January 1998 to 16 September 2020. With the analysing of LSTM, they generated the daily prices of several financial indicators. They state that the LSTM offer a way to capture diversity characteristics of financial market and detect the time-series interdependencies between observations. Importantly, it can optimise the decision of financial trading for future.

Furthermore, in the hydrological field, Choi, et al (2022) explored how to upgrade the input data to leverage the performance of LSTM. They collected data from various factors of meteorology, and integrated data from all available basins to examine if it can improve the performance of LSTM. After analyzing different type of input data, they stated that converting various variables into an input variable can significantly facilitate the performance of LSTM. In this study, it can be confirmed that clean and valued input data can positively improve the output quality of LSTM.

Because of the specification of the LSTM, it also is implemented in medical area. Zhao, et al (2023) conducted a method to detect the classification of the pan-cancer under the LSTM applied. For model processing, preprocessing the RNA-Seq data from various public sources was crucial for analyzing. They stated that LSTM neural network has the potential to detect data that are not included in the training dataset.

Reviewing these studies,LSTM can solve realistic problems greatly,because it is sentient to time-series data across various dataset,and it can offer a valued mechanism to remember the important information for a long time.It can lead to accelerating the time of processing massive data.

2.6 Tools of Data Visualization

Data visualization can easily convey hard,dry and difficult content to tangible,friendly-understanding and transparency slides.In this project,it adopts ‘matplotlib.pyplot’ as the the visualization tool during the processing.

Matplotlib is a powerful library of visualization,specially in python world.Moreover,it also offers a variety of dashboard like bar chart,column chart,pie chart and area chart,which can provide a broaden tools to demonstrate the different stories from users.At end of processing,it exports a meaningful and tangible report.

2.7 Summary

Overall,this chapter demonstrates that current proposed various literatures developed models which are from hybrid regression and outlier detection methods to advanced deep learning approaches to predict the the price of the exchange rate .More various macroeconomics data can significantly improve the accuracy of the prediction of LSTM,and plays an important role in model variable during the analyzing of model.

Conveying the hard analysis to tangible insight, Matplotlib displays a vital ability in data visualization tools.It offers a real-time view during the operation.

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Introduction

This section demonstrates how to do successful research for forecasting the exchange rate price of currency,CNY against USD with the implement of LSTM.For ensuring the success and stability of this study,it is essential to apply the data lifecycle framework ,CRISP-DM.Furthermore,this section overviews the detail of every step during applying the CRISP-DM.In the end of this section,they are discussed ,including how to set suitable metrics for evaluating the model and relative tools applied.

Aligning with the framework of CRISP-DM,fully employing crucial six steps enhances the scalability and well-performance of the model.This framework covers Business Understanding,Data Understanding,Data preparation,modelling,evaluation and deployment.

3.2 Project Framework

3.2.1 Data Science Lifecycle Framework

This methodology was developed in later 1990s by a sets of companies(Daimler-Benz, SPSS, NCR) ,however,nowadays,it is still widely applied around the field.As Wirth , et al,(2000) statement,if you want to increase the success of a data mining project,CRISP-DM is your best choice.It is consisted by four different levels of abstraction,which are phase, generic task, specialized task, and process instance.Also it has six different steps,which includes Business

understanding,Data understanding,Data Preparation,Modeling,Evaluation and Deployment.The below figure 3.1 shows the diagram of CRISP-DM.

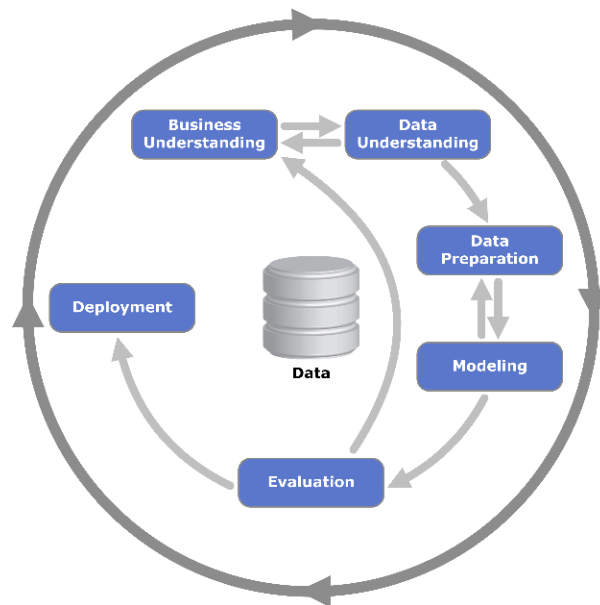


Figure 3.1 Six phrases of the CRISP-DM model (Wirth and Hipp, 2000)

3.2.2 Briefing Project Framework

This stage usually states the details of the target problem, while it is essential to set a specific goal to accomplish, and design the metric for quantifying the result and performance of the model. In this project, for accurately predicting the exchange rate of currency pairs, CNY against USD, it is crucial to apply every step of the framework, which displays on below Figure 3.2.

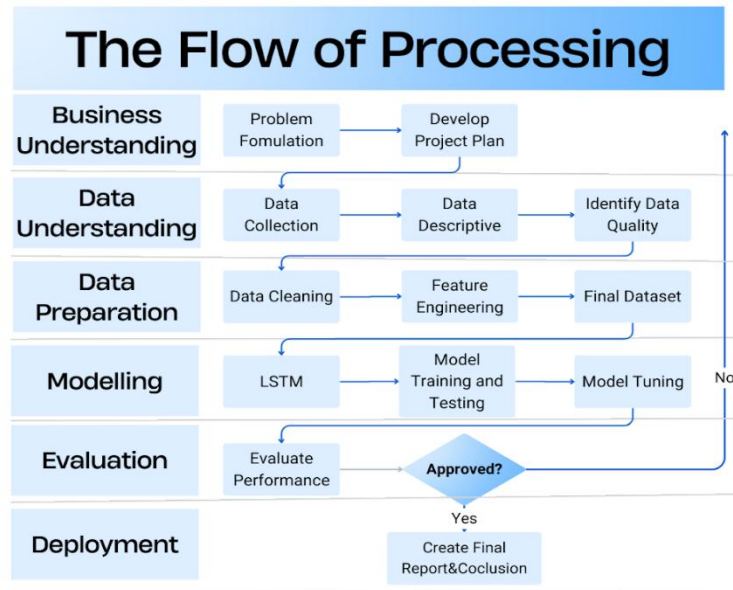


Figure 3.2 the Workflow of Machine Learning Model for Forecasting the Currency Pairs CNY/USD

3.3 Business Understanding

FX markets store substantial amounts of wealth opportunity, while involving high-Beta system crises at the same time. Because there are potential risks during trading different currencies in an enormous FX market for the stakeholders including central banks, transnation enterprises and individuals, developing a valued and accurate prescription is a key to decrease loss. Furthermore, effective techniques or relative methods can mitigate the concerns from stakeholders.

In this study, it mainly focuses on the future trending of USD against CNY. Analysing the historical exchange rate of USD/CNY to offer a suggesting pattern, and considering the impact of macroeconomic indicators, including interest, CPI and long-term government debt. Later, adopting the LSTM offers a strong power to process these data in order to predict future values.

To address these concerns, the aim of this project is to address forecasting the value of the exchange rate between CNY and USD, meantime, improving the accuracy of the predicting model.

3.3.1 Literature Review

Currently, there are lots of extraordinary papers across the world. For gaining a better understanding of the deep learning methods, these outstanding studies can provide a crucial reference for this research. To consider the influence of different economic indicators, reviewing these relative papers can build a confidence for electing indicators for this project.

Specially, choosing the effective machine learning algorithms, relative deep learning research resources can bolster the benchmark of the model from this project. Referring to LSTM resources, it can lead to successful processing of this study by employing modelling.

3.3.2 Planning and Initial Study

This stage is the main foundation of this study, and it sets the major tone for the whole research. Apparently, it aligns with the phase of CRISP-DM :Business understanding. Moreover, this stage is to erect a clear comprehensive to tackle the objective of this research. The goal of this study is to predict the exchange rate of currency pairs CNY against USD by using Long Short-Term Memory (LSTM) neural networks while capturing long-term dependencies in time-series daily data .

Going through the relative literature which studying forex prediction and LSTM applied, it provides an insight into how to facilitate model accuracy, how to predict the forex, how to apply LSTM in realistic problems, and how to leverage the

ability of LSTM. It also offers a perspective to research the forex prediction implementation, and demonstrate the limitations of every method.

For ensuring the accuracy of the model, setting a metric system is quite essential. The accuracy impacts on the success of the project, normally RMSE, MAE and MAPE as evaluation metrics are quite suitable to adopt across the model for measuring the performance of the model.

Furthermore, there is also a need to be concerned about the resources used in the project. For the programming environments, it majorly depends on the python programming. For the data resource, it derives from central bank statistic dataset and Alpha Vantage.

The output of this stage is to accomplish an initial and stable framework of solving the objective. Every mission of the stage completed contributes to operate the later stages smoothly.

3.4 Data Understanding

In this project, the phase of Data Understanding demonstrates how to collect raw data, explore data and identify the quality of collecting data.

To address the target, it is important to retrieve data from reliable sources like official government websites and high-credibility websites of finance. Also, it can be noticed that picking effective data features plays an important role in researching. In this study, historical exchange rates (USD/CNY), interest, CPI and long government debt from both sides (China and U.S). Aligning the requirement of modelling, the datetime setting is from 1, Jan 2014 to 31, Dec 2024. 11-year dataset can provide confidence of model training and testing.

After retrieving the raw data,exploring the deep meaning of the data contributes to building a stable fundament for later analysing.Using ‘plt’ from python and Matplotlib tools can offer a straightforward and tangible first-impression for raw data.

Using python methods to identify the quality of raw datasets,it suggests the security of later data cleaning.The situation of datasets decides how to employ strategies of data cleaning in the later processing.

3.5 Data Preparation

This phase covers the part of data cleaning and feature engineering,and also is the third step of the CRISP-DM framework.It is a crucial basement for modelling and later analysing.

In the part of data cleaning,the aim of this project is to provide a final dataset for model training and testing,which meets the requirements of next-phase operation.Based on real-time specialities of datasets,which are structured datasets from central banks and financial platforms,the strategy of this part should be cleaning every small dataset,then integrating small datasets into a big datasets.Finally,the big dataset contains key features under the same datetime.

Analysing the correlation among different features,python package tools offer a wonderful power to make the target come true. Understanding the relationship among several features can strengthen the processing of modelling-phase.

3.6 Modeling

The modeling training phase concentrates on the construction and training of the LSTM algorithm architecture. This phase covers model training, testing and tuning, and it dedicates to outputting the final model for predicating.

It starts from the input layer which receives sequential time-series data, and then it is followed by one or more LSTM layers designed to capture temporal dependencies. It outputs temporal patterns of LSTM layer extracted. Later, extra dense layers are created before the final output layer, which produces the predicted exchange rate of currency pairs CNY/USD. Dense layer can transform high dimensional variables into more useful representations for prediction

After the dense layer, it goes through the final layer. This layer exports the final exchange rates of currency pairs CNY/USD which LSTM generates from the learning patterns. It conveys a single actionable number (the forecasted forex value) to the front end.

3.7 Evaluation

The target of this phase is to assess the performance of the model framework with the considerations of some metrics. Normally, it sets some crucial measurements to describe the ability of the model, LSTM.

In this study, the performance of the prediction model is measured by metrics: RMSE, MAE, and MAPE, which offers a comprehensive perspective of the accuracy. Root Mean Squared Error (RMSE) strengthens the large errors. And Mean Absolute Error (MAE) displays straightforward interpretation of average error. As for Mean Absolute Percentage Error (MAPE) illustrates the percentage of accuracy.

For checking the error if it is random, constructing Residual Analysis is the way to examine the randomness. It also validates if it is truly useful in real-world trading. Furthermore, testing data from 1 January, 2020 to 31 December, 2022 specially is a key step to validate if LSTM can generate the exchange rate accurately without known condition in extreme period.

3.7.1 Root Mean Squared Error

RMSE stands for Root Mean Squared Error, and it measures the square root of the average squared the difference between predicted and actual values. It is suitable for assessing large error. The below figure 3.3 shows the formula of RMSE.

$$RMSE = \sqrt{\sum_{i=1}^n \frac{(\hat{y}_i - y_i)^2}{n}}$$

Figure 3.3 the Formula of RMSE

3.7.2 Mean Absolute Error

MAE stands for Mean Absolute Error, and it measures the average absolute difference between predicted and actual values. It is easy to demonstrate the outliers without substantial sensitivity. The below figure 3.4 shows the formula of MAE.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Figure 3.4 The Formula of MAE

3.7.3 Mean Absolute Percentage Error

MAPE stands for Mean Absolute Percentage Error, and it illustrates the error as a percentage of the real forex value. It is straightforward to display the total accuracy of this model. The below figure 3.5 shows the formula of MAPE.

$$MAPE = \frac{1}{n} \sum_{t=1}^n \left| \frac{V(t) - P(t)}{V(t)} * 100 \right|$$

Figure 3.5 The Formula of MAPE

3.8 Deployment

While the final model meets the requirements of business, it is time to deploy the model to the platforms which are decided by stakeholders, such as central banks, transnational enterprises and individuals. Meantime, it should be noticed that monitoring the performance of the model over the time also is a key measurement.

Due to the objectives of this study, deploying the model is out of range about researching. It does not demonstrate how to deploy the model on the system.

CHAPTER 4

BUSINESS AND DATA UNDERSTANDING

4.1 Introduction

This chapter majorly illustrates the details of how to collect data and clean data, how to process the data with the application of LSTM, how to evaluate the model, and how to visualize the final result with an accurate model. The steps are designed by 5 different steps to explain and demonstrate. From dataset overview, dataset Cleaning and preparation, EDA, and performance of LSTM to visualization of the result, all steps initial display the programme of processing.

4.2 Business Understanding

4.2.1 Objective of Business

In this project, it aims to address forecasting the exchange rate (USD/CNY) with regard to the macroeconomic indicators, including CPI, interest and long-term government debt.

4.2.2 Translation of Goals

To tackle this objective effectively, it is essential to convert this concern to use a model to forecast the exchange rate. Due to the specificity of the problem, employing the LSTM model can solve concerns well.

4.2.3 Evaluation metrics

For assessing the success of the final result, it is important to establish some quantitative metrics to verify whether the model solution is successful or not. Because this is a non-classification question, adopting MAE, RMSE and MAPE, these three metrics can be suitable for testing the quality of result from output of model.

4.3 Data Understanding

4.3.1 Initial Data Collection

Because of involving two countries and 4 different features, this study built 7 different datasets to store valued data at the beginning. Moreover, for U.S side relative data, it is retrieved from Alpha Vantage website and International Monetary Fund, by the contrast, Chinese side is collected from the source which includes the Central Bank of the People's Republic of China and International Monetary Fund.

'df_FOREX_original' is labeled by storing the historical exchange rate (USD/CNY) from 04-21-2006 to 12-15-2025. It also can be viewed by figure 4.1. The figure 4.2 displays that the size of this dataset is 5000 rows and 5 columns. And there is no null cell from 5 different columns, which indicates 'timestamp', 'open', 'high', 'low', 'low' and 'close', while the data type of 'timestamp' is objective, and the other 4 are float64. For initial analyzing, figure 4.3 can tell the value of two columns from the sides of mean, min, max, 25%, 50%, 75%, std.

	timestamp	open	high	low	close
0	2025-12-15	7.0555	7.0565	7.0463	7.0480
1	2025-12-12	7.0560	7.0579	7.0535	7.0546
2	2025-12-11	7.0595	7.0642	7.0550	7.0580
3	2025-12-10	7.0655	7.0700	7.0598	7.0646
4	2025-12-09	7.0720	7.0733	7.0631	7.0636
...	...	...	...	...	...
4995	2006-04-27	8.0180	8.0189	8.0145	8.0162
4996	2006-04-26	8.0200	8.0204	8.0124	8.0163
4997	2006-04-25	8.0144	8.0145	8.0095	8.0095
4998	2006-04-24	8.0169	8.0170	8.0130	8.0163
4999	2006-04-21	8.0185	8.0194	8.0110	8.0145

Figure 4.1 The historical Data of Exchange Rate (USD/CNY)

```

RangeIndex: 5000 entries, 0 to 4999
Data columns (total 5 columns):
#   Column      Non-Null Count  Dtype
---  -
0   timestamp   5000 non-null   object
1   open        5000 non-null   float64
2   high        5000 non-null   float64
3   low         5000 non-null   float64
4   close       5000 non-null   float64
dtypes: float64(4), object(1)
memory usage: 195.4+ KB
None

```

Figure 4.2 The Basic Information of 'df_FOREX_original'

	open	high	low	close
count	5000.000000	5000.000000	5000.000000	5000.000000
mean	6.778933	6.786288	6.771762	6.778969
std	0.450005	0.450064	0.449634	0.449602
min	6.042600	6.043700	6.040800	6.041200
25%	6.387075	6.393150	6.381725	6.387075
50%	6.808700	6.818350	6.799200	6.809050
75%	7.075250	7.086525	7.065700	7.075550
max	8.025900	8.030000	8.022500	8.030000

Figure 4.3 The Descriptive Statistics of 'df_FOREX_original'

'df_10Y_US2' is stored the long-term government debt rate of U.S from 02-01-1962 to 12-12-2025. It also can be viewed by figure 4.4. For initial analysing, because the data type of 'value' was objective, it is necessary to convert objective to float at first. The figure 4.5 shows that the size of this dataset is 16684 rows and 2 columns. And in the 'Date' column there is no a null cell, whereas the 'value' column presents 712 cells with null notification, while the data type of 'timestamp' is objective, and 'value' is float64. For initial descriptive statistics, the figure 4.6 can tell the value of two columns from the sides of mean, min, max, 25%, 50%, 75%, std.

	timestamp	value
0	2025-12-12	4.19
1	2025-12-11	4.14
2	2025-12-10	4.13
3	2025-12-09	4.18
4	2025-12-08	4.17
...	...	...
16679	1962-01-08	4.03
16680	1962-01-05	4.02
16681	1962-01-04	3.99
16682	1962-01-03	4.03
16683	1962-01-02	4.06

Figure 4.4 The Historical Data of Long-term Government Debt Rate in U.S


```

RangeIndex: 16684 entries, 0 to 16683
Data columns (total 2 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   timestamp   16684 non-null  object
 1   value       15972 non-null  float64
dtypes: float64(1), object(1)
memory usage: 260.8+ KB
None

```

Figure 4.5 The Basic Information of 'df_10Y_US2'

```

              value
count  15972.000000
mean      5.821563
std       2.941967
min       0.520000
25%       3.880000
50%       5.460000
75%       7.550000
max      15.840000

```

Figure 4.6 The Descriptive Statistics of 'df_10Y_US2'

'df_cpi_us' is stored the consumer price indicator (CPI) of U.S ,which covers from 01-01-2014 to 01-11-2024.It also can be viewed by figure 4.7.The figure 4.8 presents that the size of this dataset is 131 rows and 2 columns.And there is no any null cell in two columns,while the data type of 'Date' is datetime,and 'CPI_US' is float64.For initial descriptive statistics,the figure 4.9 can describe the value of two columns from the sides of mean,min,max,25%,50%,75%,std.

	Date	CPI_US
0	2024-11-01	144.684725
1	2024-10-01	144.763145
2	2024-09-01	144.596674
3	2024-08-01	144.365081
4	2024-07-01	144.247680
..	...	...
126	2014-05-01	109.100665
127	2014-04-01	108.720945
128	2014-03-01	108.363696
129	2014-02-01	107.670295
130	2014-01-01	107.273607

Figure 4.7 The Historical Consumer Price Indicators(CPI) in U.S

```

RangeIndex: 131 entries, 0 to 130
Data columns (total 2 columns):
 #   Column  Non-Null Count  Dtype
---  -
 0   Date    131 non-null     datetime64[ns]
 1   CPI_US  131 non-null     float64
dtypes: datetime64[ns](1), float64(1)
memory usage: 2.2 KB
None

```

Figure 4.8 The Basic Information of 'df_cpi_us'

count	131.000000
mean	120.993218
std	12.090772
min	107.177760
25%	110.701404
50%	117.466883
75%	130.980416

Figure 4.9 The Descriptive Statistics of 'df_cpi_us'

'df_interest_US1' is interest of U.S ,which covers from 01-07-1954 to 01-11-2025.It also can be viewed by figure 4.10.The figure 4.11 presents that the size of this dataset is 857 rows and 2 columns.And there is no any null cell in two columns,while the data type of 'timestamp' is object,and 'CPI_US' is float64.For initial descriptive statistics,the figure 4.12 can describe the value of two columns from the sides of mean,min,max,25%,50%,75%,std.

	timestamp	value
0	2025-11-01	3.88
1	2025-10-01	4.09
2	2025-09-01	4.22
3	2025-08-01	4.33
4	2025-07-01	4.33
..	...	...
852	1954-11-01	0.83
853	1954-10-01	0.85
854	1954-09-01	1.07
855	1954-08-01	1.22
856	1954-07-01	0.80

Figure 4.10 The Historical Interest in U.S

```

RangeIndex: 857 entries, 0 to 856
Data columns (total 2 columns):
#   Column      Non-Null Count  Dtype
---  -
0   timestamp    857 non-null    object
1   value        857 non-null    float64
dtypes: float64(1), object(1)
memory usage: 13.5+ KB
None

```

Figure 4.11 The Basic Information of ‘df_interest_US1’

```

              value
count  857.000000
mean    4.604831
std     3.544852
min     0.050000
25%     1.910000
50%     4.330000
75%     6.130000
max     19.100000

```

Figure 4.12 The Descriptive Statistics of ‘df_interest_US1’

‘df_cpi_china’ is the dataset to save data from the consumer price indicator(CPI) from China.It consists of two different columns and 132 rows,which it is displayed in figure 4.13.It also describes the data type of two columns,which shows that the data type of ‘Date’ is datetime and ‘CPI_China’ is float64 .From figure 4.14 to see,it starts from 01-01-2014 to 01-12-2024.Moreover,the figure 4.15 presents the descriptive of ‘CPI_China’,including count,mean,standard deviation,minimum ,maxmum,25%,50%,and 75% .

```

RangeIndex: 132 entries, 0 to 131
Data columns (total 2 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   Date        132 non-null    datetime64[ns]
 1   CPI_China   132 non-null    float64
dtypes: datetime64[ns](1), float64(1)
memory usage: 2.2 KB
None

```

Figure 4.13 The Basic Information of 'df_cpi_china'

```

      Date    CPI_China
0  2024-12-01  132.036865
1  2024-11-01  132.036865
2  2024-10-01  132.806012
3  2024-09-01  133.062394
4  2024-08-01  133.062394
..      ...      ...
127 2014-05-01  112.968447
128 2014-04-01  112.853524
129 2014-03-01  113.313217
130 2014-02-01  113.887821
131 2014-01-01  113.313217

```

Figure 4.14 The Historical Consumer Price Indicator (CPI) 'df_cpi_china'

```

count    132.000000
mean     124.118962
std       6.965540
min      112.853524
25%      117.680253
50%      124.230807
75%      131.460005
max      133.575159

```

Figure 4.15 The Descriptive Statistics of 'df_cpi_china'

‘df_interest_China_original’ is the dataset to save Chinese interest data from 02-01-2014 to 20-12-2024. In raw dataset, it is a .xlsx file and there are 11 different sheets in one excel file. These 11 excels share same format, and include 1 year interest rate and 5 years interest rate. The figure 4.16 illustrates the part of Chinese historical interest rate in 2023. And the figure 4.17 suggests that there are 3 different columns with 12 rows, and no null cells. It also tells that the data type of ‘Date’ is object. For initial analysing, it converts the data type of ‘1Y’ and ‘5Y’ from object to floats in advance. As for the figure 4.18 provides the basic descriptive statistics of ‘df_interest_China_original’ in 2023. The table 4.1 shows the entire raw dataset initial information.

	Date	1Y	5Y
0	20 Dec 2023	3.45	4.2
1	20 Nov 2023	3.45	4.2
2	20 Oct 2023	3.45	4.2
3	20 Sep 2023	3.45	4.2
4	21 Aug 2023	3.45	4.2
5	20 Jul 2023	3.55	4.2
6	20 Jun 2023	3.55	4.2
7	22 May 2023	3.65	4.3
8	20 Apr 2023	3.65	4.3
9	20 Mar 2023	3.65	4.3
10	20 Feb 2023	3.65	4.3
11	20 Jan 2023	3.65	4.3

Figure 4.16 The Part of Historical Interest in China

```

RangeIndex: 12 entries, 0 to 11
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  -
 0   Date    12 non-null     object
 1   1Y       12 non-null     float64
 2   5Y       12 non-null     float64
dtypes: float64(2), object(1)
memory usage: 420.0+ bytes
None

```

Figure 4.17 The Part Basic Information of ‘df_interest_China_original’

	1Y	5Y
count	12.000000	12.000000
mean	3.550000	4.241667
std	0.095346	0.051493
min	3.450000	4.200000
25%	3.450000	4.200000
50%	3.550000	4.200000
75%	3.650000	4.300000
max	3.650000	4.300000

Figure 4.18 The Descriptive Statistics of ‘df_interest_China_original’ in 2023

File Name	Description
Sheet0	Year:2023 Columns:'Date','1Y','5Y' Row: 12
Sheet0(2)	Year:2022 Columns:'Date','1Y','5Y' Row: 12
Sheet0(3)	Year:2021 Columns:'Date','1Y','5Y' Row: 12
Sheet0(4)	Year:2020 Columns:'Date','1Y','5Y' Row: 12
Sheet0(5)	Year:2019 Columns:'Date','1Y','5Y' Row: 12
Sheet0(6)	Year:2018 Columns:'Date','1Y','5Y' Row: 12
Sheet0(7)	Year:2017 Columns:'Date','1Y','5Y' Row: 12
Sheet0(8)	Year:2016 Columns:'Date','1Y','5Y' Row: 12
Sheet0(9)	Year:2015 Columns:'Date','1Y','5Y' Row: 12
Sheet0(10)	Year:2014 Columns:'Date','1Y','5Y' Row: 12
Sheet0(11)	Year:2024 Columns:'Date','1Y','5Y' Row: 12

Table 4.1 the Whole Information of the ‘df_interest_China_original’

‘df_10Y_China_Original’ is the dataset to save Chinese long-term government debt rate data from 02-01-2014 to 27-12-2024. In raw dataset, it is a .xlsx file and there are 11 different sheets in one excel file. These 11 excels share same format, and include 10 years government debt rate. Because this project only focus on 10 years government debt rate of China, the raw dataset retrieves same time data from Chinese central bank website. Figure 4.19 displays the situation of Chinese long-term government debt rate in 2014. And the figure 4.20 suggests that there are 10 different columns and 247 rows. It is straightforward to see that from third column to eighth column, and the final column share empty data. As for the figure 4.21 provides the basic descriptive statistics of ‘df_10Y_China_Original’ in 2014. The table 4.2 presents the whole raw dataset initial information.

	曲线名称	日期	3月	6月	1年	3年	5年	7年	10年	30年
0	中债国债收益率曲线	2014-12-26	NaN	NaN	NaN	NaN	NaN	NaN	3.6339	NaN
1	中债国债收益率曲线	2014-12-25	NaN	NaN	NaN	NaN	NaN	NaN	3.6305	NaN
2	中债国债收益率曲线	2014-12-24	NaN	NaN	NaN	NaN	NaN	NaN	3.5975	NaN
3	中债国债收益率曲线	2014-12-23	NaN	NaN	NaN	NaN	NaN	NaN	3.6604	NaN
4	中债国债收益率曲线	2014-12-22	NaN	NaN	NaN	NaN	NaN	NaN	3.7030	NaN
..	...	...	..	..	..	..	..	...	...	
242	中债国债收益率曲线	2014-01-08	NaN	NaN	NaN	NaN	NaN	NaN	4.5919	NaN
243	中债国债收益率曲线	2014-01-07	NaN	NaN	NaN	NaN	NaN	NaN	4.6420	NaN
244	中债国债收益率曲线	2014-01-06	NaN	NaN	NaN	NaN	NaN	NaN	4.6569	NaN
245	中债国债收益率曲线	2014-01-03	NaN	NaN	NaN	NaN	NaN	NaN	4.6415	NaN
246	中债国债收益率曲线	2014-01-02	NaN	NaN	NaN	NaN	NaN	NaN	4.6018	NaN

Figure 4.19 the 2014’s Long-term government Debt Rate of China


```

RangeIndex: 247 entries, 0 to 246
Data columns (total 10 columns):
#   Column      Non-Null Count  Dtype
---  -
0   曲线名称    247 non-null   object
1   日期        247 non-null   object
2   3月         0 non-null     float64
3   6月         0 non-null     float64
4   1年         0 non-null     float64
5   3年         0 non-null     float64
6   5年         0 non-null     float64
7   7年         0 non-null     float64
8   10年        247 non-null   float64
9   30年        0 non-null     float64
dtypes: float64(8), object(2)

```

Figure 4.20 The Basic Information of ‘df_10Y_China_Original’ in 2014

```

      3月   6月   1年   3年   5年   7年      10年  30年
count  0.0   0.0   0.0   0.0   0.0   0.0  247.000000   0.0
mean   NaN   NaN   NaN   NaN   NaN   NaN    4.162231   NaN
std    NaN   NaN   NaN   NaN   NaN   NaN    0.308920   NaN
min    NaN   NaN   NaN   NaN   NaN   NaN    3.476700   NaN
25%    NaN   NaN   NaN   NaN   NaN   NaN    4.005850   NaN
50%    NaN   NaN   NaN   NaN   NaN   NaN    4.231000   NaN
75%    NaN   NaN   NaN   NaN   NaN   NaN    4.454600   NaN
max    NaN   NaN   NaN   NaN   NaN   NaN    4.656900   NaN

```

Figure 4.21 The Descriptive Statistics of ‘df_10Y_China_Original’ in 2014

File Name	Description
国债及其他债券收益率曲线	Year:2014 Column:‘曲线名称’,‘日期’,‘3月’,‘6月’,‘1年’,‘3年’,‘5年’,‘7年’,‘10年’,‘30年’ Row:247
国债及其他债券收益率曲线 (2)	Year:2015 Column:‘曲线名称’,‘日期’,‘3月’,‘6月’,‘1年’,‘3年’,‘5年’,‘7年’,‘10年’,‘30年’ Row:247

国债及其他债券收益率曲线 (3)	Year:2016 Column:'曲线名称','日期','3月','6月','1年','3年','5年','7年','10年','30年' Row:247
国债及其他债券收益率曲线 (4)	Year:2017 Column:'曲线名称','日期','3月','6月','1年','3年','5年','7年','10年','30年' Row:247
国债及其他债券收益率曲线 (5)	Year:2018 Column:'曲线名称','日期','3月','6月','1年','3年','5年','7年','10年','30年' Row:247
国债及其他债券收益率曲线 (6)	Year:2019 Column:'曲线名称','日期','3月','6月','1年','3年','5年','7年','10年','30年' Row:247
国债及其他债券收益率曲线 (7)	Year:2020 Column:'曲线名称','日期','3月','6月','1年','3年','5年','7年','10年','30年' Row:247
国债及其他债券收益率曲线 (8)	Year:2021 Column:'曲线名称','日期','3月','6月','1年','3年','5年','7年','10年','30年' Row:247
国债及其他债券收益率曲线 (9)	Year:2022 Column:'曲线名称','日期','3月','6月','1年','3年','5年','7年','10年','30年' Row:247
国债及其他债券收益率曲线 (10)	Year:2023 Column:'曲线名称','日期','3月','6月','1年','3年','5年','7年','10年','30年' Row:247
国债及其他债券收益率曲线 (11)	Year:2024 Column:'曲线名称','日期','3月','6月','1年','3年','5年','7年','10年','30年' Row:247

Table 4.2 the Whole Information of the 'df_10Y_China_Original

CHAPTER 5

DATA PREPARATION AND EDA

5.1 Introduction

This chapter majorly covers the part of data cleaning and initial exploratory data analysis from cleaned data. It specifically presents details of these two parts with clear processing. In data cleaning part, it pays attention to select data features and period, remove data duplication, null cells and other noises, then incorporate seven different datasets into one comprehensive dataset, finally transform data outface to be aligned with the requirement of modelling.

On the side of exploratory data analysis, it suggests digging out the relationship of six features with the target feature, FX exchange rate (USD/CNY). Moreover, it offers an initial analysis of the seven features to find historical patterns

5.2 Data Cleaning

5.2.1 Data Selection

Through the phase of Data Understanding processing, it gains seven small different datasets with different periods and columns. For addressing the objective of the Business Understanding, it is a crucial step to pick out the essential features and a specific period. Meeting the requirements of forecast the exchange rate between USD and CNY, strict picking can significantly boost the later processing and modelling.

On the period side, it is decided to adopt the time from 01-01-2014 to 31-12-2024. It covers total 11 years historical data, which provides a reasonable time for

analysing,specifically including the COVID19 period.It also can tell enormous social crisis whether impact the trending of exchange rate (USD/CNY) or not.

As for the side of features,it considers exchange rate,which is the target feature,interest,CPI,and long-term government debt rate from China and U.S both sides.With the help of six important and effective features and the target ,it enhances the effectiveness of the model processing.

5.2.2 Cleaning Data

This sub-node illustrates the cleaning seven small datasets procedure,and at end of time,it outputs a cleaned and easy-analyzing dataset for later processing.There are seven small datasets,including ‘df_FOREX_original’,‘df_10Y_US2’,‘df_cpi_us’, ‘df_interest_US1’, ‘df_cpi_china’,‘df_interest_China_original’,and ‘df_10Y_China_Original’,which stands for storing the historical exchange rate,and interest ,CPI and long-term government debt rate from both sides.

Observing the initial information of ‘df_FOREX_original’,it faces two major problems in order to align with the requirements of final datasets.Checking on ‘timestamp’ column,it needs to firstly remove the extra data rows ,which exceed the range of the period from 01-01-2014 to 31-12-2024,while converting the data type from object to datetime,and secondly,fill the missing value by using ‘.ffill()’.Taking look on ‘open’,‘high’,‘low’.and ‘close’ columns,in this study ,it is decided to save ‘close ’ value as the daily exchange rate.The final cleaning side of ‘df_FOREX_original’ shows on figure 5.1,while it is named by ‘df_FOREX’.

	Date	USD_rate_CNY
0	2024-12-31	7.2994
1	2024-12-30	7.2994
2	2024-12-27	7.2950
3	2024-12-26	7.2975
4	2024-12-25	7.2940
...	...	...
2778	2014-01-07	6.0512
2779	2014-01-06	6.0526
2780	2014-01-03	6.0515
2781	2014-01-02	6.0507
2782	2014-01-01	6.0540

Figure 5.1 The Cleaning Dataset Review of 'df_FOREX'

Taking serious consideration on 'df_10Y_US2', it is obvious that this dataset need to make a cleaning on columns 'timestamp' and 'value'. For 'timestamp', firstly keeping the range of date is under the range of target of period from 01-01-2014 to 31-12-2014, meantime, converts data type from object to datetime. As for the column 'value', it only need to fill all null cell. The figure 5.2 displays the final cleaning dataset, which is called by 'df_10Y_US'.

	Date	10YTreasury_US
2	2024-12-29	4.62
3	2024-12-28	4.62
4	2024-12-27	4.62
5	2024-12-26	4.58
6	2024-12-25	4.58
...	...	...
4013	2014-01-05	3.01
4014	2014-01-04	3.01
4015	2014-01-03	3.01
4016	2014-01-02	3.00
4017	2014-01-01	3.00

Figure 5.2 The Cleaning Dataset Review of 'df_10Y_US'

Looking at the dataset 'df_cpi_us', because the original source provides a structured dataset, in this phase, there is no cleaning part to employ it. The figure 5.3 presents the final cleaning dataset view in the current phase.

	Date	CPI_US
0	2024-11-01	144.684725
1	2024-10-01	144.763145
2	2024-09-01	144.596674
3	2024-08-01	144.365081
4	2024-07-01	144.247680
..	...	...
126	2014-05-01	109.100665
127	2014-04-01	108.720945
128	2014-03-01	108.363696
129	2014-02-01	107.670295
130	2014-01-01	107.273607

Figure 5.3 The Cleaning Dataset Review of 'df_cpi_us'

Review on the dataset 'df_interest_US1', it meets one problem to clean. Column 'timestamp' is out of the target date range, and adjusting the range aligning with the period from 01-01-2014 to 31-12-2025 is a wise move. Then it has to convert data type of the 'timestamp' from object to datetime. The figure 5.4 displays the final cleaning dataset, named by 'df_interest_US'.

	Date	interest_US
0	2024-12-01	4.48
1	2024-11-30	4.64
2	2024-11-29	4.64
3	2024-11-28	4.64
4	2024-11-27	4.64
...	...	...
3983	2014-01-05	0.07
3984	2014-01-04	0.07
3985	2014-01-03	0.07
3986	2014-01-02	0.07
3987	2014-01-01	0.07

Figure 5.4 The Cleaning Dataset Review of 'df_interest_US1'

Due to the providing of the structure data , 'df_cpi_china' does not need to make a movement in the current phase. The figure 5.5 illustrates the view of the final cleaning dataset of 'df_cpi_china'.

	Date	CPI_China
0	2024-12-01	132.036865
1	2024-11-01	132.036865
2	2024-10-01	132.806012
3	2024-09-01	133.062394
4	2024-08-01	133.062394
..	...	...
127	2014-05-01	112.968447
128	2014-04-01	112.853524
129	2014-03-01	113.313217
130	2014-02-01	113.887821
131	2014-01-01	113.313217

Figure 5.5 The Cleaning Dataset Review of 'df_cpi_china'

Through a series observation,the dataset ‘df_interest_China_original’ faces multiple procedure to clean and adjust.Because different yearly data store in the different sheets of excel,first of all,it is necessary to load all sheets into one dataframe,and secondly,removing the noisy data like non-valued string information and transform Chinese version to English version for name of columns.In this study ,it is discussing the short interest rate,so saving the column ‘1Y ’ is an accurate movement.Then,for keeping alignment of requirement,converting data type of ‘date’ from object to datetime is an essential movement.The figure 5.7 presents the view of the final cleaning dataset of ‘df_interest_China_original’,which is newly named by ‘df_interest_China’.

	Date	interest_China
0	2024-12-20	3.10
1	2024-11-20	3.10
2	2024-10-21	3.10
3	2024-09-20	3.35
4	2024-08-20	3.35
...	...	...
1467	2014-01-08	5.73
1468	2014-01-07	5.73
1469	2014-01-06	5.73
1470	2014-01-03	5.73
1471	2014-01-02	5.73

Figure 5.6 The Cleaning Dataset Review of ‘df_interest_China’

For the Chinese long-term government rate dataset ‘df_10Y_China_Original’ ,it meets a similar situation with the ‘df_interest_China’.Because daily data stores among 11 sheets of excel,firstly,it is a better way to load all sheet into one dataframe of pandas,and then secondly removing noisy and non-valued string,while translating Chinese version to English version among all names of columns.Finally,it has to remove 100% empty columns.While dealing with the ‘Date’,it needs to convert the data type from objective to datetime.The figure 5.7 illustrates the look of the final cleaning dataset,named by ‘df_10Y_China’ .

	Date	10Y_China
0	2024-12-27	1.6929
1	2024-12-26	1.7122
2	2024-12-25	1.7365
3	2024-12-24	1.7166
4	2024-12-23	1.7019
...	...	...
2746	2014-01-08	4.5919
2747	2014-01-07	4.6420
2748	2014-01-06	4.6569
2749	2014-01-03	4.6415

Figure 5.7 The Cleaning Dataset Review of 'df_10Y_China'

5.2.3 Integrating Dataset and Data Normalisation

Through the early series of processing, it can be obtained that seven small cleaned datasets. However, it still can not meet the dataset requirements of the model. For making true those regulation, it is a necessary to integrate the seven datasets into one single dataset with clear date range from 01-01-2014 to 31-12-2024, while it does not miss any date.

By using the python method, combining seven datasets into one single dataset, named by 'df_whole_frame' with non-missing date time from 01-01-2014 to 31-12-2024. It is straightforward to find there are many null cells among seven features. Because the authority of the government department only report monthly data for CPI and interest. For aligning with daily time series, it has to change CPI and interest data from monthly to daily while fill null cells with the data of last day. Then as for long-term government rate, and exchange rate, they are daily change, so they only need to fill null cells with data of last day. The figure 5.8 shows the view of final cleaning

dataset,named by ‘df_whole_frame’.Moreover,the figure 5.9 presents the descriptive statistics of the ‘df_whole_frame’,and the figure 5.10 illustrates the basic information of dataframe, ‘df_whole_frame’.

The ‘df_whole_frame’ is suitable for later modeling phase after cleaning processing,and it can support the model testing,model training and model tuning,which can be considered as an effective fundament.

	USD_rate_CNY	10YTreasury_US	CPI_US	CPI_China	interest_US	\
Date						
2014-01-01	6.0540	3.00	107.273607	113.313217	0.07	
2014-01-02	6.0507	3.00	107.273607	113.313217	0.07	
2014-01-03	6.0515	3.01	107.273607	113.313217	0.07	
2014-01-04	6.0515	3.01	107.273607	113.313217	0.07	
2014-01-05	6.0515	3.01	107.273607	113.313217	0.07	
...	...	...	...	...	...	
2024-12-27	7.2950	4.62	144.684725	132.036865	4.48	
2024-12-28	7.2950	4.62	144.684725	132.036865	4.48	
2024-12-29	7.2950	4.62	144.684725	132.036865	4.48	
2024-12-30	7.2994	4.62	144.684725	132.036865	4.48	
2024-12-31	7.2994	4.62	144.684725	132.036865	4.48	

	interest_China	10Y_China
Date		
2014-01-01	5.73	4.60
2014-01-02	5.73	4.60
2014-01-03	5.73	4.64
2014-01-04	5.73	4.64
2014-01-05	5.73	4.64
...	...	...
2024-12-27	3.10	1.69
2024-12-28	3.10	1.69
2024-12-29	3.10	1.69
2024-12-30	3.10	1.69
2024-12-31	3.10	1.69

Figure 5.8 The Historical Cleaning Data of ‘df_whole_frame’

	USD_rate_CNY	10YTreasury_US	CPI_US	CPI_China	interest_US \
count	4022.000000	4022.000000	4022.000000	4022.000000	4022.000000
mean	6.702182	2.486363	121.170703	124.109282	1.628834
std	0.346126	1.004274	12.179279	6.942263	1.844476
min	6.041200	0.520000	107.177760	112.853524	0.050000
25%	6.394975	1.780000	110.684207	117.680253	0.100000
50%	6.711950	2.360000	117.523291	124.230807	0.845000
75%	6.973200	2.980000	132.401040	131.588196	2.390000
max	7.345000	4.980000	144.763145	133.575159	5.330000

	interest_China	10Y_China
count	4022.000000	4022.000000
mean	4.200144	3.132277
std	0.654749	0.538024
min	3.100000	1.690000
25%	3.700000	2.780000
50%	4.300000	3.090000
75%	4.310000	3.500000
max	5.770000	4.660000

Figure 5.9 The Descriptive Statistics of 'df_whole_frame'

```
DatetimeIndex: 4022 entries, 2014-01-01 to 2024-12-31
Data columns (total 7 columns):
#   Column          Non-Null Count  Dtype
---  -
0   USD_rate_CNY     4022 non-null   float64
1   10YTreasury_US   4022 non-null   float64
2   CPI_US           4022 non-null   float64
3   CPI_China        4022 non-null   float64
4   interest_US      4022 non-null   float64
5   interest_China   4022 non-null   float64
6   10Y_China        4022 non-null   float64
dtypes: float64(7)
memory usage: 251.4 KB

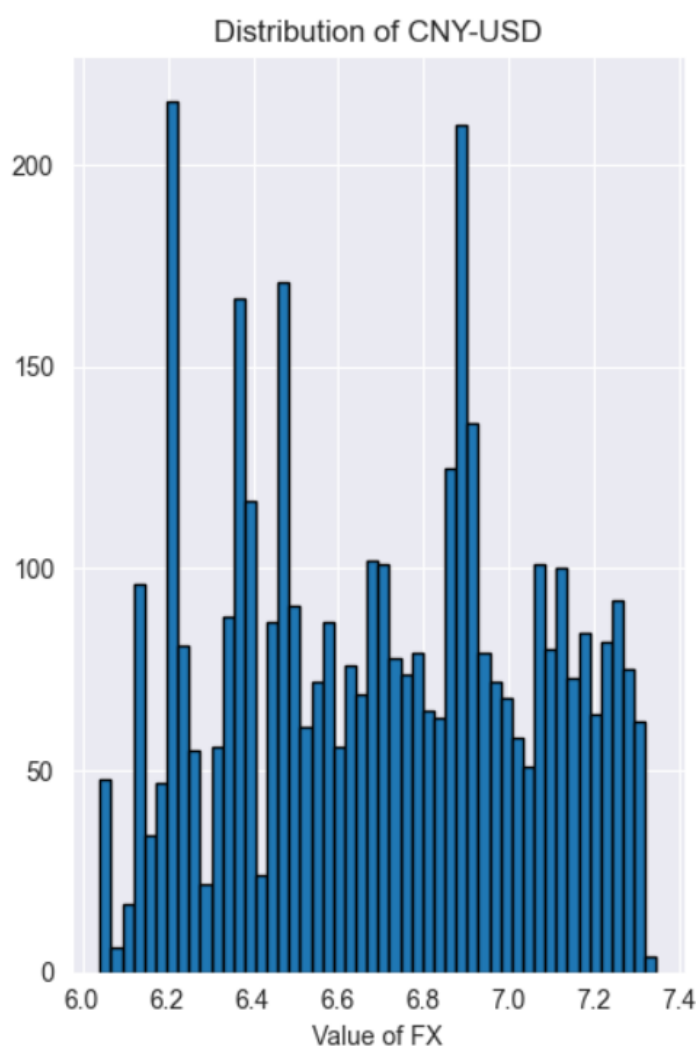
None
```

Figure 5.10 The Basic Information of 'df_whole_frame'

5.3 Exploratory Data Analysis

5.3.1 Initial Quality of Dataset

Through the analysis of the boxplot about the target feature, column 'USD_rate_CNY', which the figure 5.11 presents, the result shows that the skewness is -0.02, which it is extreme close to 0 and suggests that the distribution of 'USD_rate_CNY' is symmetric without extreme left or right skew. The Kurtosis equals -1.15, which it reflects that it is a flatter distribution than normal distribution without any outlier.



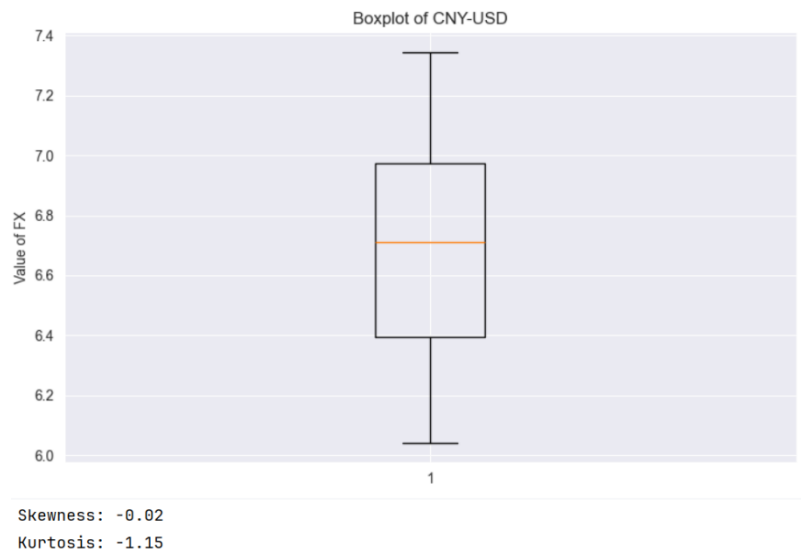
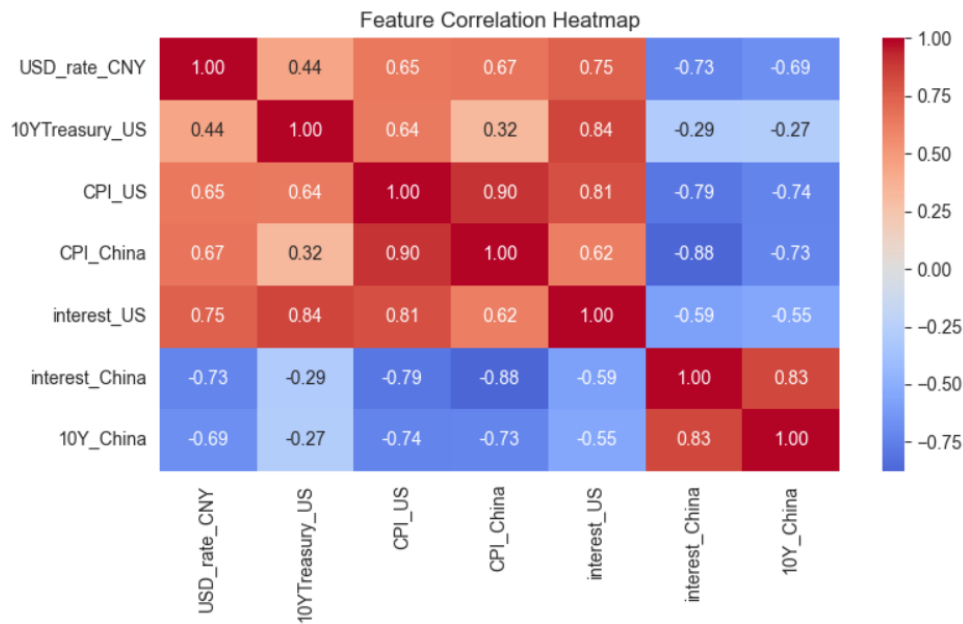


Figure 5.11 The Distribution of Target Feature in Boxplot

5.3.2 Correlation of Features

For viewing the relationship between target feature and the six features whether it is strong or not, plotting a Correlation Heatmap can clearly to see the situation. It can be told that the target feature, 'USD_rate_CNY' has strongest correlated with 'interest_US'. It said that the major sentiment of influencing the exchange rate (USD/CNY) is interest rate of U.S, and it totally corresponds to the real-life regulations, which refers to USD occupy dominance position in the world. It also can find that '10Y_China' and 'interest_China' presents negative correlation with the 'USD_rate_CNY', which indicates that high Chinese interest rate leads to strong CNY, comparing with the U.S interest. On the other side, it conformed to the basic macroeconomic theory: high interest policy leads to enhance the power of domestic currency.



Features most correlated with USD_rate_CNY:

```
USD_rate_CNY      1.000000
interest_US       0.745388
CPI_China         0.666632
CPI_US            0.652608
10YTreasury_US    0.437177
10Y_China         -0.687099
interest_China    -0.728708
Name: USD_rate_CNY, dtype: float64
```

Figure 5.12 The Correlation Review of Target Feature

5.3.3 Trending of Features

The figure 5.13 displays the trending change of exchange rate (USD/CNY) from 2014 to 2024, and it can be observed that around Feb 2018 and Feb 2022, the line has an extreme turning-downward, which implies strong CNY and weak USD. Generally to see, the trending turn upward over the time, which it indicates that USD shows strong position, comparing with CNY.

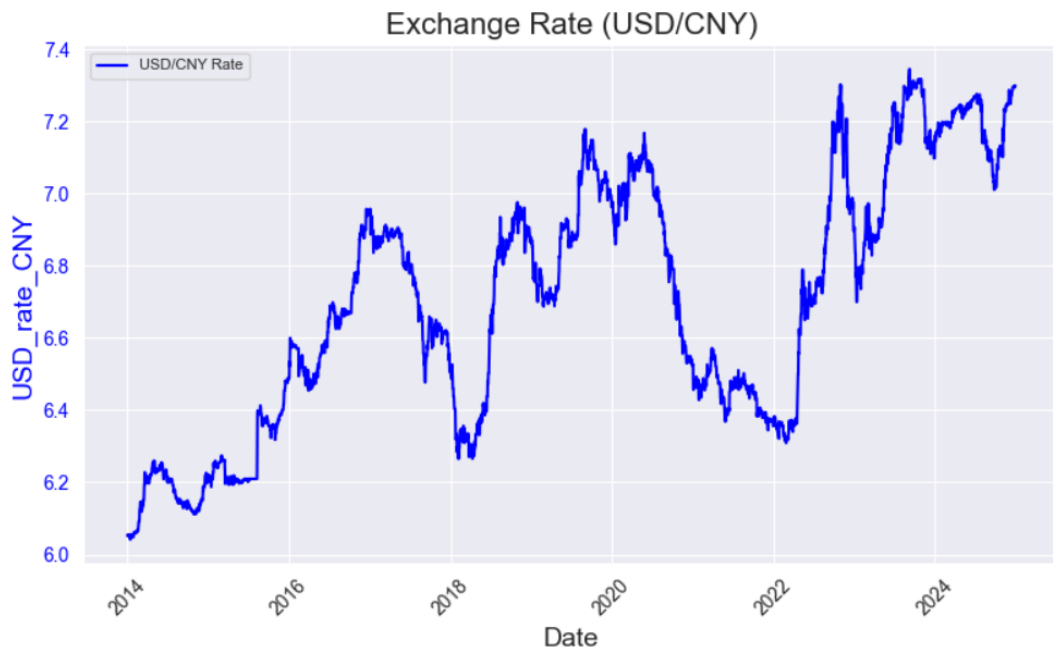


Figure 5.13 The Trending of Exchange Rate (USD/CNY) over the Target Years

When considering the interest influence, it will be able to demonstrate the the trending of pairs USD/CNY. From the Figure 5.14 to see, it is so clear to figure out the explanation from 2020 to 2022 . Because the Chinese interest was much higher than the US's which led to CNY gets strong increasing contrasting with USD , and the market expected Federal Reserve would decrease the future interest rate, the trending of USD/CNY intends to show declining over the years.

On the other hand, from 2023 to 2024, due to the continually boosting interest rate by the Federal Reserve tightening policy, the trend started to turn upward from the bottommost point. It is straightward to note the interest rate of US was higher than the Chinese one by almost 2% in a short time. Through the whole period, it illustrated interest played a significant role in the movement of the FOREX directly.



Figure 5.14 Review the Trending of Interest Rate and FX(USD/CNY)

Figure 5.15 illustrates the fluctuation of CPI and exchange rate through the period of 2014 until 2024. Generally, both curves are intending to turn upward in long time because central banks need to maintain the 2% CPI goal to ensure domestic economy runs healthily. From 2020 and 2024, the CPI of US presents consistent upward movement, suggesting an increasing in value. On the other hand, Chinese one presents quite stable and slow increasing. From 2020 to 2024, it illustrates almost flat curve, comparing with US's one. Observing the details of Chinese CPI, it may make policy maker intending to lower the interest for avoiding the risk of deflation.



Figure 5.15 Review the Trending of CPI and FX(USD/CNY)

Through Figure 5.16, it presents the trending of the exchange rate under the influence of a 10-Year bond around the 10 years. From 2014 to 2024, the curve of exchange rate shares similar direction with the curve of US ten years national bond. At the China side, over the observed period, it generally illustrates the steadily decline. Specially, from 2021 to 2024, there is a noticeable and consistent downward trend. It can assume that market expects Chinese central bank will possibly continually execute decline interest rate policy in the future.

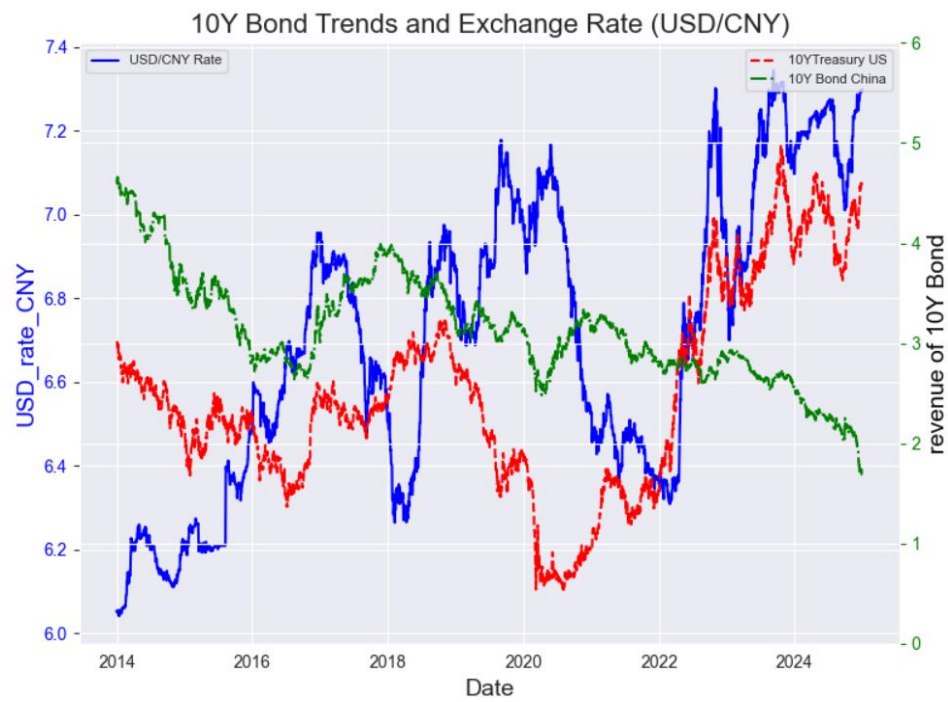


Figure 5.16 Review the Trending of Long-term Government Debt Rate and FX(USD/CNY)

CHAPTER 6

MODELLING AND EVALUATION

6.1 Introduction

This chapter demonstrates the fourth and fifth steps of the CRISP-DM framework ,and it covers Data modelling and Model Evaluation.In the part of Data Modelling,it introduces from selection of model,testing and training of model to model tuning.Moreover,in the part of evaluation of model,it presents how to assess the quality of the training model ,and check it whether meet the requirements of business objective or not.

6.2 Data Modelling

6.2.1 Model Selection

For model selection,in this study ,it confirms the LSTM as a priority choice.There are some important variances to concern.

First of all,LSTM can tackle the vanishing Gradient Problem.Looking at the dataset,it conveys multiple rows ,and the quantity of the dataset.It requires stability strictly for high-accurate model prediction.LSTM use memory cells and gates to offer a security way to allow the model processing stably.

Second,in this study,it needs to find effective patterns of learning from seven features and 11 years period.It suggests a highly well-performing model to discover the relationship between past and future from historic data,specially for time-series

data.LSTM provides an excellent power for improving the accuracy of forecasting through remembering patterns from past,even far.

Thirdly,LSTM also can decide individually what to remember ,what to forget,and what to output.It can facilitate the accuracy of model prediction further.

Finally,with these outstanding specialties of LSTM,which refers to well-solving on vanishing gradient problems,efficient-learning from long dependencies ,and stably-selecting memory from gates,it contributes to significantly improving the accurate value of predicting,and transforming raw dataset to high-accurate forecasting future value.

6.2.2 Model Test and Train

First of all,it has to set a target column,which is 'USD_rate_CNY'.Because LSTM only accepts the scaled data,it needs to transform 'df_whole_frame' to scaled status.Then making a sequences of LSTM store the scaled data,target columns and setting of time_steps,which it refers to how many past data or observations are used to predict next value.

In this study,it conducts the strategy ,which is 80% data of 'df_whole_frame' for training,and 20% data for testing.The figure 6.1 shows the size of training and testing datasets.After splitting the dataset for training and testing,it is time to make a LSTM model to train the model with series setting.

During this period of training model,it consider setting the value of epochs as 50,and the value of 'batch_size' as 32.After processing,the figure 5.2 displays the movement of model loss and MAE during the model training.In the two charts,it can be seen that for train loss and train MAE,it starts increasing at first,then around 15th epochs,and it shows a significant drop.After that,it is goes down to around 0. However,for validation loss and validation MAE,it beginning with quickly increasing

,and then there is an obvious drop around 15th epochs.But it can see an sudden increase and drop around 35th epochs.Finally,it turns down to around 0.

X_train shape: (3169, 60, 7)

y_train shape: (3169,)

X_test shape: (793, 60, 7)

y_test shape: (793,)

Figure 6.1 The Size of Training and Testing Datasets



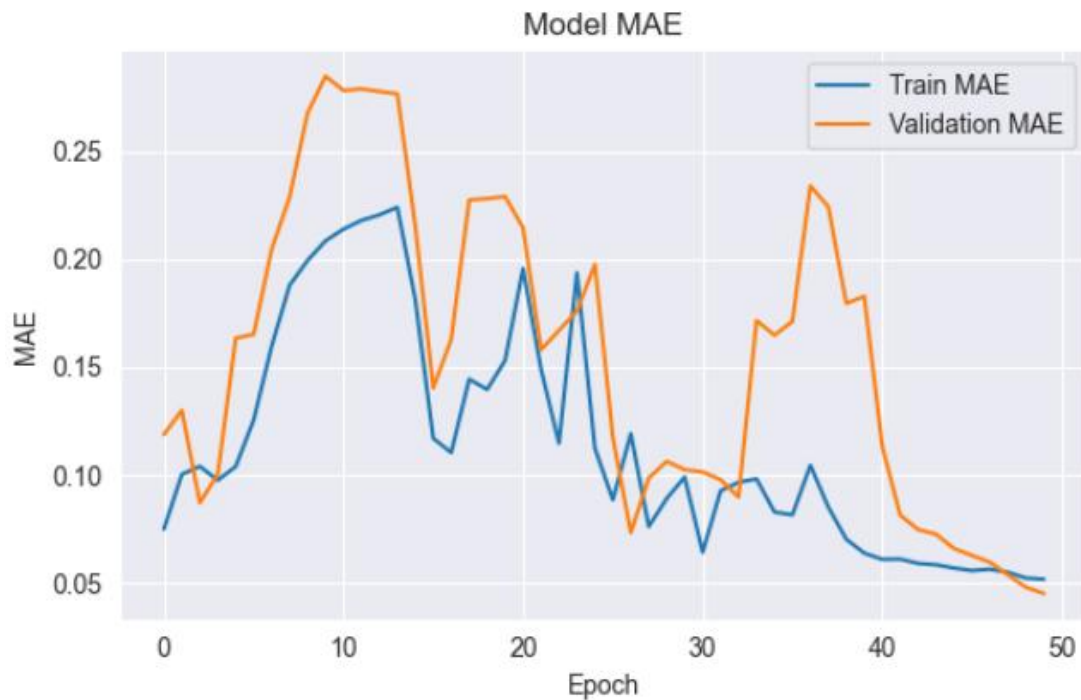


Figure 6.2 the Trending of Model Loss and MAE over Epochs

Through observing Figure 6.3, it presents the trending of actual exchange rate compared with predicted exchange rate. It can be told that at first, the predicted curve line is close to the actual curve line, then it shows a significant gap between the actual one and the predicted one. And at end of time, it changes close to the actual one. It suggests the movement can align with the result of Figure 6.2, which are the chart of Model Loss and Model MAE.

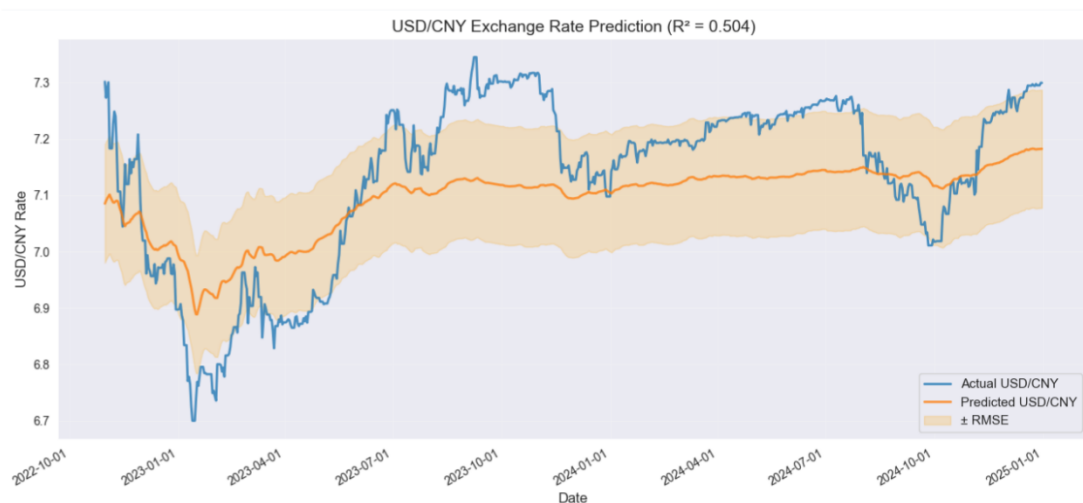


Figure 6.3 The Trending of Actual VS Predicted Exchange Rate (USD/CNY)

For figuring out the specific situation of the training model over time, residuals can offer a detailed view to analyse. Residual stands for the subtraction between actual value and predicted value, which refer to 'true_scaled - pred_scaled'. Red line means perfect predicted, on the other hand, points above 0 refer to the model is under the predicted compared to points below 0 reflects the model is over the predicted. Analysing with three different charts, it suggests that most errors stand at around '0.1'. For gaining a well-performed [model.it](#) needs to make sure the most points is near the 0.



Figure 6.4 The Review of Actual VS Predicted Residuals

Figure 6.5 reflects the quality of the first training model, which indicates that the value of R square equals 0.5043, MAE is 0.0791, RMSE is 0.0903, and MAPE shows 8.57%. 0.5043 of R square value means that this training model performed moderately to predict target value, but not strongly. However, from the side of MAE and RMSE, it can tell that the error is small, suggesting that this training model can be conformed as a moderate model. Because of high value of MAPE, it implies this model is not suitable for final well-performance training model to deploy.

R2 Score: 0.5043

MAE: 0.0791

RMSE: 0.0903

MAPE: 8.57%

Figure 6.5 The Quality View of first Training and Testing

6.2.3 Model Tuning

For facilitating the performances,it is necessary to use a strategy of model tuning.In this case,decreasing ‘batch_size ‘ to 8,it can produce a better performance theoretically.So it has to set ‘batch_size ’ from 32 to 8,while keeping other parameters unchanged.For running procedure,it still same with first model training.

After processing with adjusting the ‘batch_size’,Figure 6.6 provides a much better result than the first model training.From both charts,it can be observed that at first,the curve line of train loss increases shortly,the turns down to around the 0.For the curve line of validation loss.it also share same pattern.Reviewing the result of two charts,it conveys a better news to check the metrics.

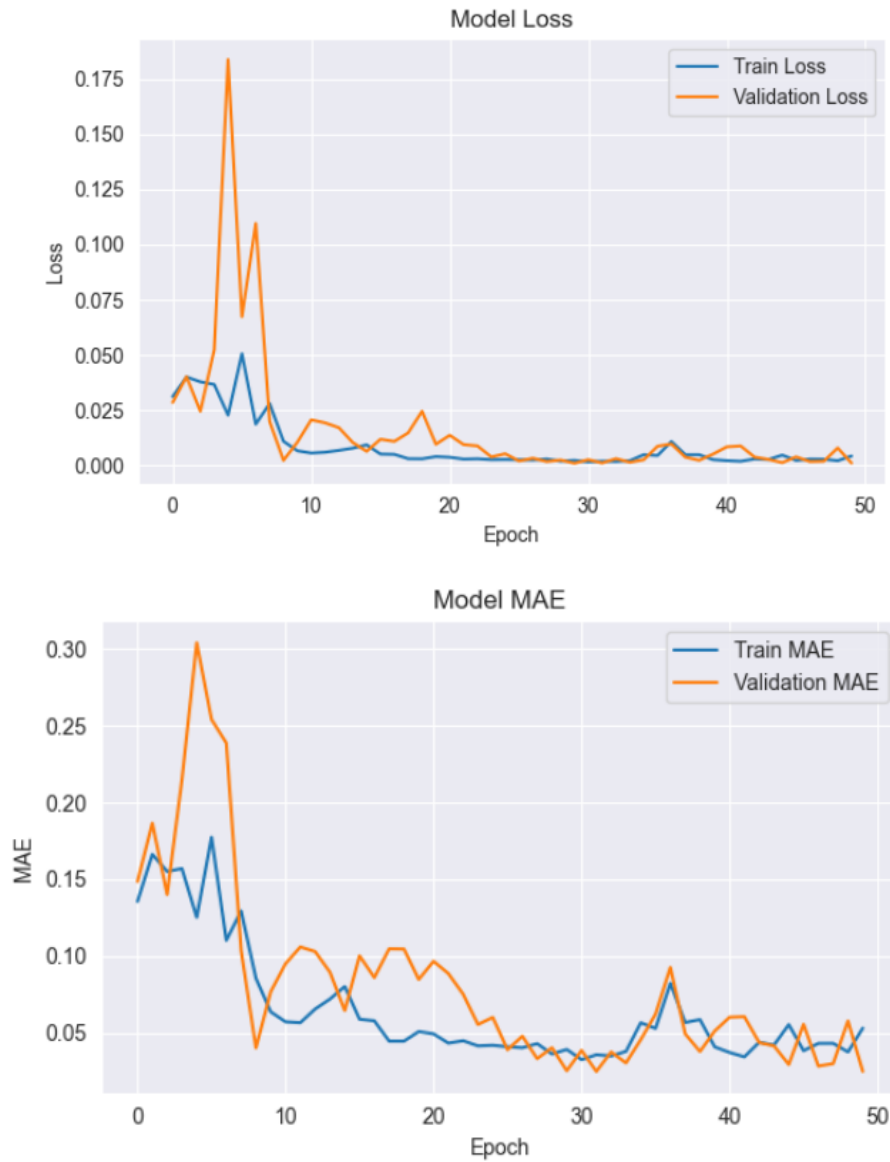


Figure 6.6 the Trending of Tuning Model Loss and MAE over Epochs

From Figure 6.5, it illustrates the predicted curve line is close to the actual curve line tightly. Compare with the first model training, tuning model can produce more accurate predicted value while performing well during testing and training.

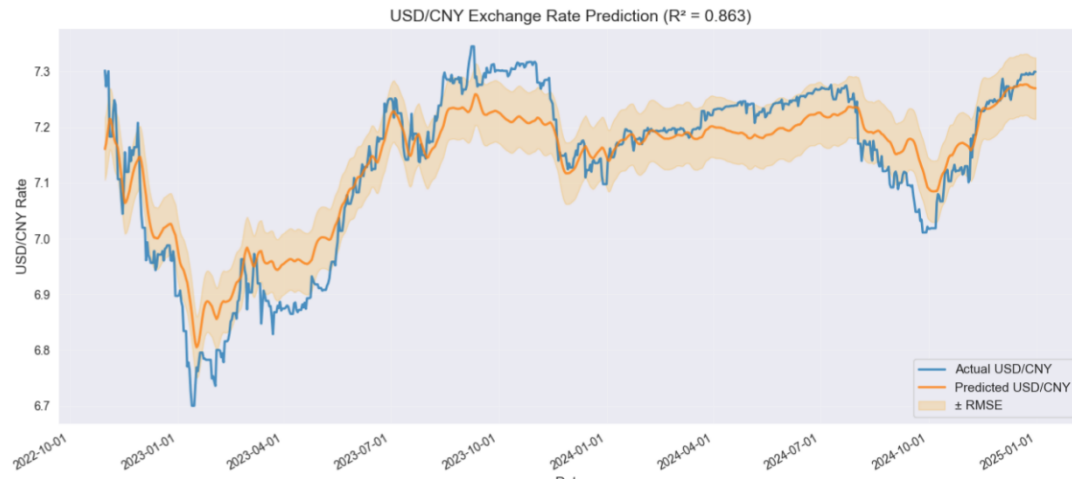
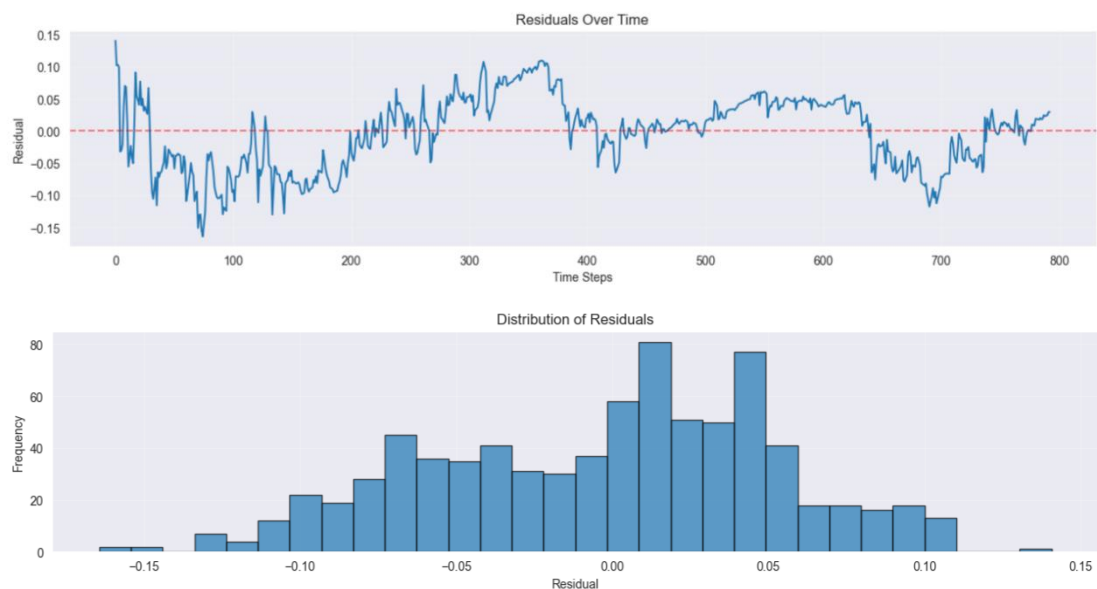


Figure 6.7 The Trending of Tuning Model Actual VS Predicted Exchange Rate (USD/CNY)

Like the first training model, it has to do a residual analysing for check the location of errors. Figure 6.6 presents the predicted errors over the time, and it can be concluded that most predicted points locate from 0 to 0.05, which conveys strong and impressive news of model performance with considering the residuals distribution of the first model training.



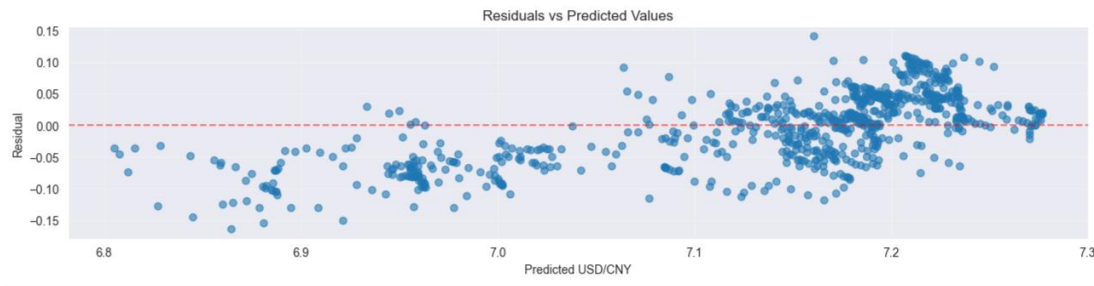


Figure 6.8 The Review of Tuning Model Actual VS Predicted Residuals

Observing Figure 6.7, it displays the evaluation metrics of tuning model, which is 0.8627 of R square value, 0.0391 of MAE value, 0.0475 of RMSE value, and 4.45% of MAPE value. High R Square means the model can explain 86.27% of variance, and it suggests a strong evidence. Low MAE and RMSE presents that there is no extreme outliers and errors are controlled well. Moreover low MAPE and less than 5%, it represents that model can be deployed for solving business objective, and reflects that this model performance excellently .

R2 Score: 0.8627

MAE: 0.0391

RMSE: 0.0475

MAPE: 4.45%

Figure 6.9 The Quality View of Tuning Model Training and Testing

Because the tuning model can produce excellent evaluation metrics, it indicates that this model is suitable for deploying while tackling the early business objective. Furthermore, it conveys a well-performing model to relative stakeholders.

6.3 Model Applying

6.3.1 Next-7 Days Forecasting

After training and tuning the model, it is time to test real-life FX market. First of all, it starts with forecasting next 7 days exchange rate (USD/CNY). Figure 6.8 shows the comparison between actual value and predicted value under 7 days, and it suggests there is still some gap between two lines.

Moreover, Figure 6.9 displays the evaluation metrics of predicted 7 days, which can be observed that the value of MAE equals 0.0623, the value of RMSE is 0.0806, and the value of MAPE is 0.85%. Through these three metrics, this model is competitive for forecasting short time exchange rate. The predicted value is acceptable.

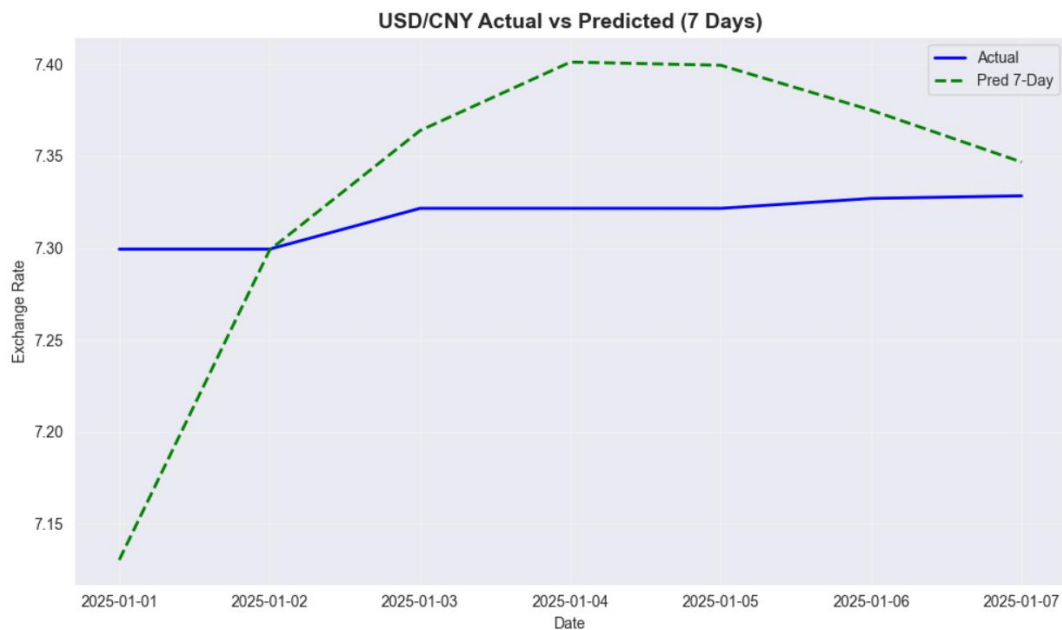


Figure 6.10 The Trending of Actual VS Predicted Next-7 Days Exchange Rate (USD/CNY)

MAE: 0.0623

RMSE: 0.0806

MAPE: 0.85%

Figure 6.11 The Quality View of Next-7 Days Predicted

6.3.2 Next-15 Days Forecasting

After viewing the result of next-7 days' predicted value,it is time to check the next-15 days situation.Figure 6.9 presents the comparison between actual value and predicted value under 15 days,and it suggests there is still some gap between two lines.However,it also can be seen that after 01-09,the gap has a significant smaller change.

Moreover,Figure 6.10 displays the evaluation metrics of predicted 15 days,which can be observed that the value of MAE equals 0.0371,the value of RMSE is 0.0565,and the value of MAPE is 0.51%.Through these three metrics,this model shows competitive for forecasting short time exchange rate.The predicted value is acceptable.

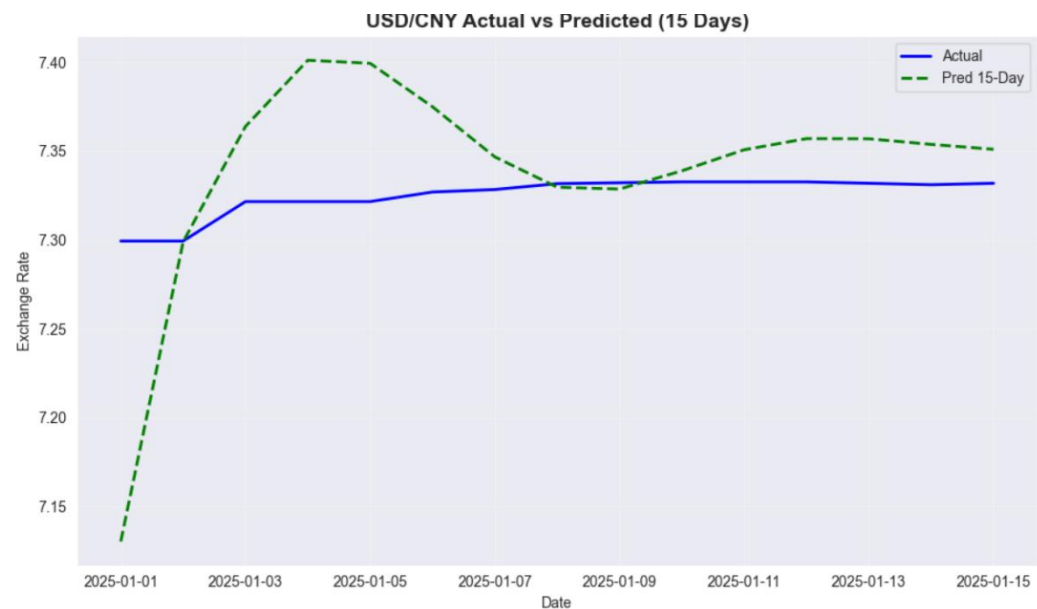


Figure 6.12 The Trending of Actual VS Predicted Next-15 Days Exchange Rate (USD/CNY)

MAE: 0.0371
RMSE: 0.0565
MAPE: 0.51%

Figure 6.10 The Quality View of Next-7 Days Predicted

6.3.3 Next-30 days Forecasting

After viewing the result of next-30 days' predicted value, it is time to check the next-30 days situation. Figure 6.11 presents the comparison between actual value and predicted value under 30 days, and it suggests there is still some gap between two lines. However, it also can be seen that after 01-17, the gap has a significant bigger change.

Moreover, Figure 6.11 displays the evaluation metrics of predicted 30 days, which can be observed that the value of MAE equals 0.0576, the value of RMSE is 0.0723, and the value of MAPE is 0.79%. Overall, these metrics suggest that the model performs well, and produces value and effective predicted values.

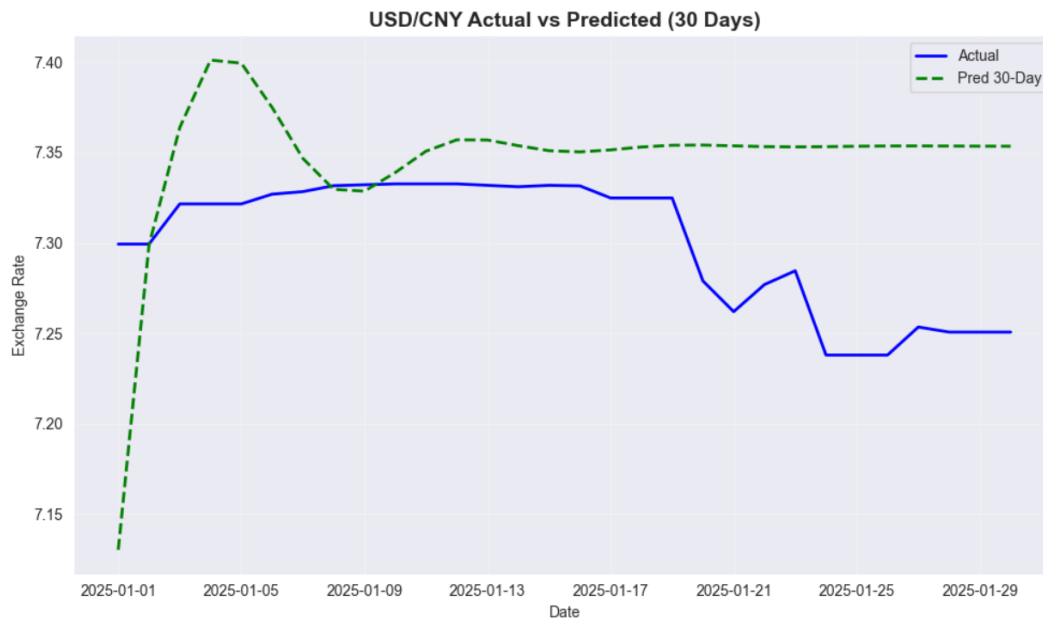


Figure 6.13 The Trending of Actual VS Predicted Next-30 Days Exchange Rate (USD/CNY)

MAE: 0.0576
 RMSE: 0.0723
 MAPE: 0.79%

Figure 6.14 The Quality View of Next-30 Days Predicted

CHAPTER 7

CONCLUSION

7.1 Introduction

In this chapter, it recaps the findings from the research, summarizes the procedure of the whole time, and concludes the significant patterns against the business objectives.

Going through the whole research, first it business understanding, data understanding, then data preparation and EDA, finally data modelling and evaluation. At the end of study, it outputs series of exchange rate values (USD/CNY) for future, which can be accepted by real-time business, while aligning the requirements from stakeholders. Checking with the metrics of assessing, it conveys a well-performed model.

7.2 Research Finding

Overall, through this study, it displays strong and excellent evidence that adjusting the values of 'epoch' and 'batch_size' of the model can facilitate the quality of the model during model training. By observing the evaluation metrics of first-training and later-training, for gaining a high-accurate model, it is essential to find perfect combination of 'epoch' and 'batch_size'.

Later,while testing the model to forecast next-7 days,next-15 days ,and next-30 days forecasting exchange rate,the model also performs quite accurately.Comparing with real-life values from 01-01-2025 to 30-01-2025 ,the forecasted values of the model producing shows the complete.

Futhermore,speaking about the finding of evaluation metrics,it is clearly to find that with the different size of ‘epoch’ and ‘batch_size’,it outputs different result on evaluation metrics,such as R square,RMSE,MAPE,and MAE.While decreasing the size of ‘epoch’ and ‘batch_size’,the result of evaluation will turn to good way.While setting the size of ‘epoch’ as 50,and the size of ‘batch_size’ as 8,it is obvious to see there is a significant improvement of evaluation metrics ,compared with old one.

Moreover,focusing on the result of Next-7 day forecasting,Next-15 day forecasting,and Next-30 day forecasting,from three evaluation metrics to consider,it can be observed that the model performs well in three different period.The evaluation metrics (MAE,MAPE,RMSE) from three different periods are close to 0,which it means model performs well.However,there is no strong and efficient evidences to support that the model performs well in short period or long period.

For those potential stakeholders like central banks,transnational enterprises,and individual investors,considering these forecasted values of the model producing may offer a successful and accurate decision-making to decrease the unmet profits loss during trading of the FX.

It also can offer a new perspective for those researchers who study FX forecasting model.

7.3 Limitations

Even though the final model already reaches the requirement of deploying for a business-level,there are some limitations of this study ,which needs to be observed.

For the model section,in this study ,it only adopts a single model to predict the future exchange rate (USD/CNY).Single model may face some challenges for processing huge datasets ,and could not master the limited evaluation metrics again.For those stakeholders who want to use it for big business,it exists some shortage of this model.

As for features,in this project,it only considers the three indicators of macroeconomic.However,there are multiple indicators which can assist the prediction .Limited features may lead to the occasional situation of the result.

In this study ,it only considers currency pairs,which is USD and CNY.It could not be proved the effectiveness of applying on other currencies pairs,such as EUR/USD,EUR/CNY,and MYR/CNY.

7.4 Future Works

Considering the above limitation,it is necessary to to improve the model scalability,accuracy and stability.For addressing these problems,later study will make a plan to meet these limitations.

Due to the three indicators of this study,it will concern more macroeconomic indicators such as PPI,GDP and so on.With the single model shortage,later study will involve at least three models to analyze the problems such as ARIMA,XGBoost and so on.With the help of compound models ,it will be suitable to process the multiple foreign exchange rates pairs.

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